Univariate stationary time series models

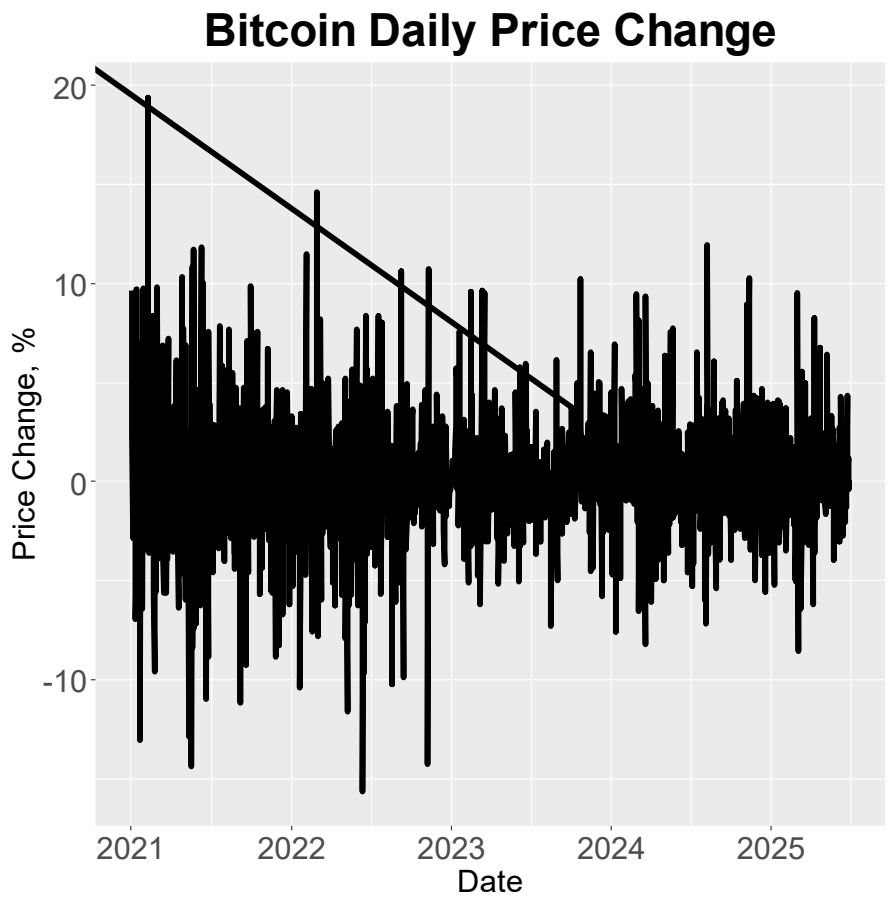
Econometrics. Lecture 8

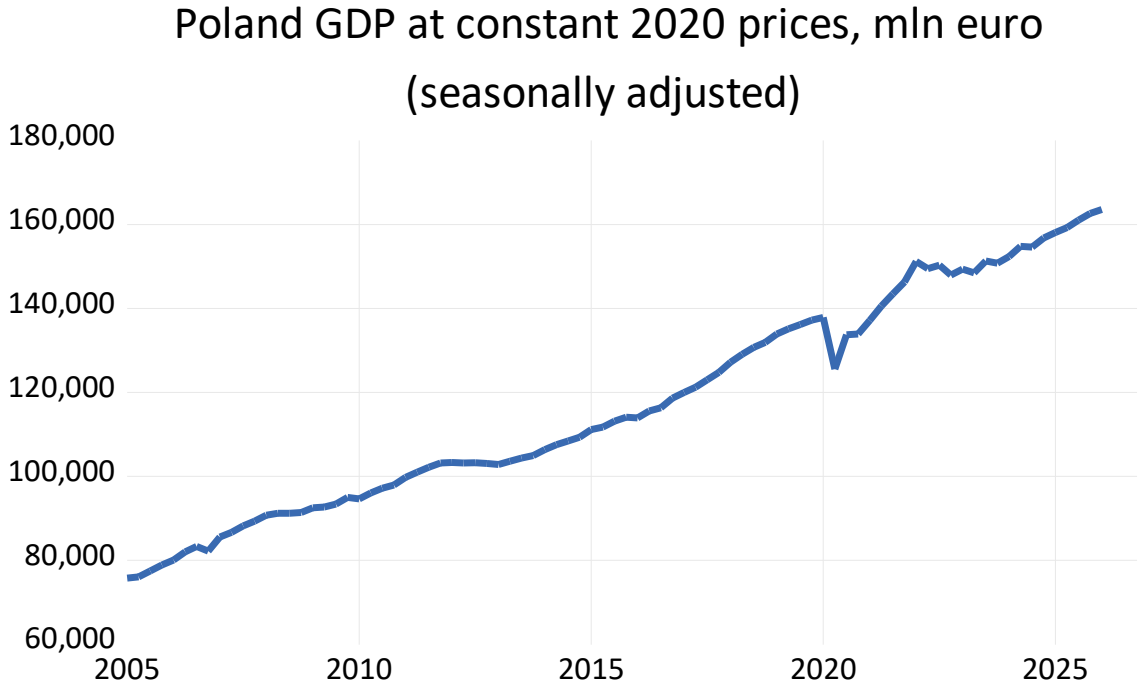
Content

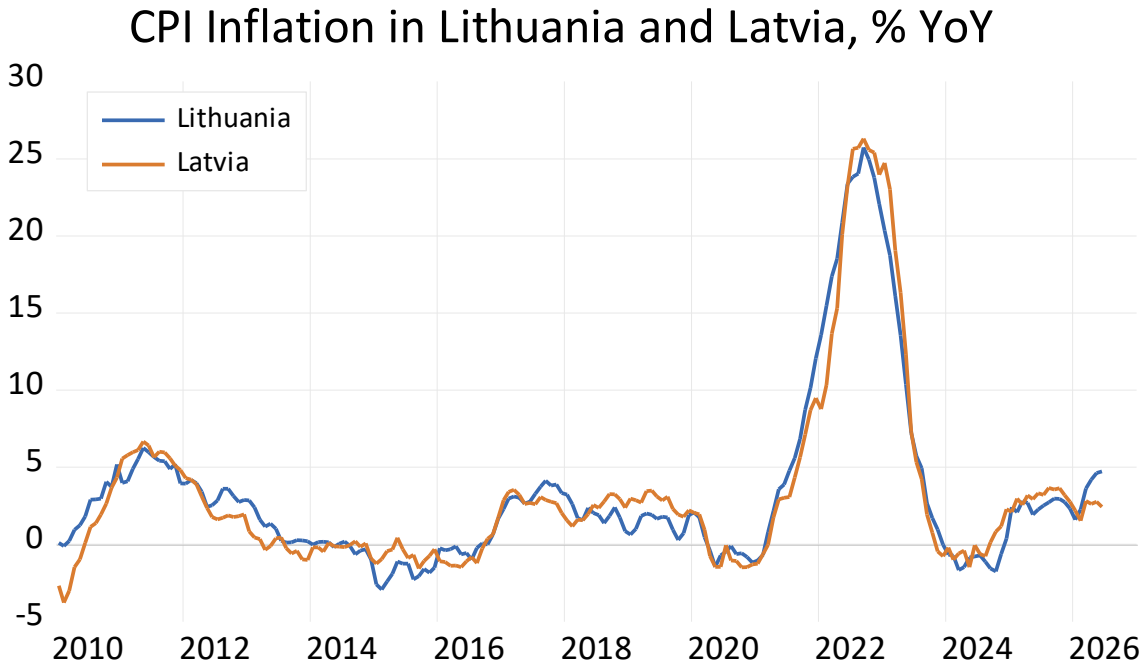
1. Time series data
2. Stationarity
3. Identification of a time series
4. Estimation and diagnostic of univariate time series models

1. Time series data

Time series

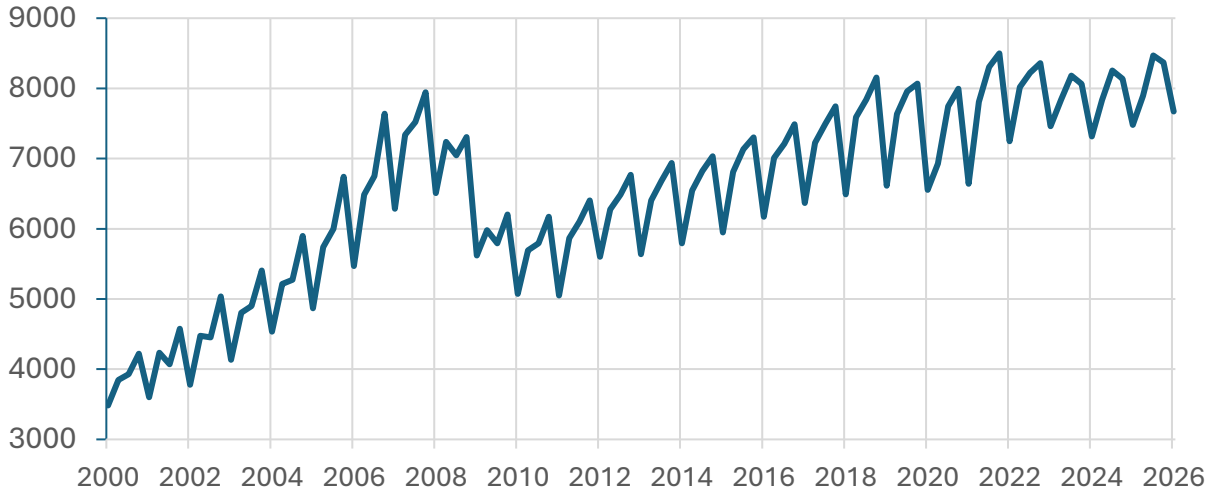

Bitcoin Daily Price Change
Price Change, %
20
10
0
-10
2021
2022
2023
2024
2025
Date


Poland GDP at constant 2020 prices, mln euro
(seasonally adjusted)
180,000
160,000
140,000
120,000
100,000
80,000
60,000
2005
2010
2015
2020
2025

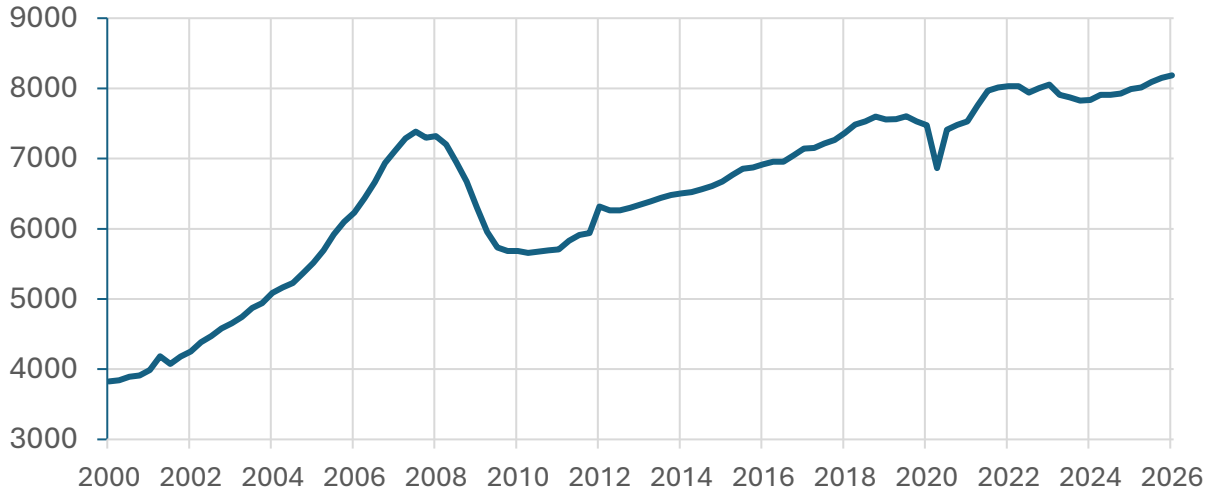

CPI Inflation in Lithuania and Latvia, % YoY
Lithuania
Latvia
30
25
20
15
10
5
0
-5
2010
2012
2014
2016
2018
2020
2022
2024
2026

Components of time series: Latvia real GDP

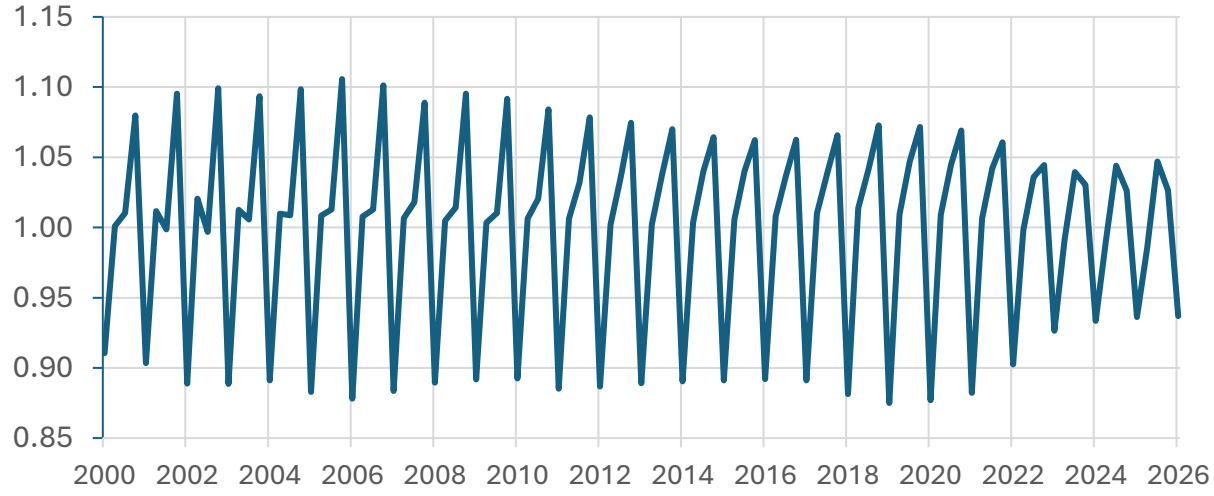
Original series (X_t)



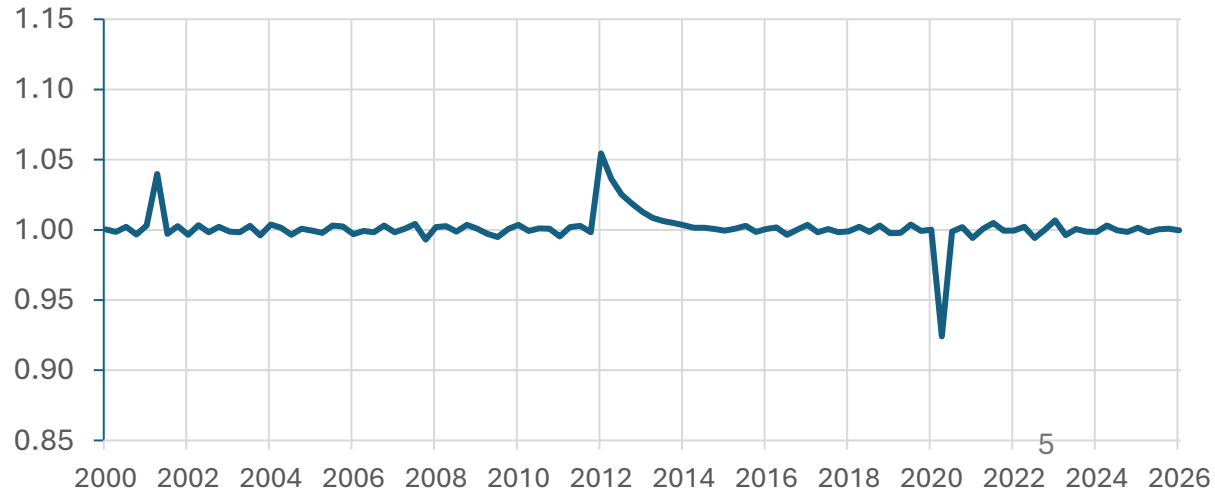
Trend-cycle component (TC_t)



Seasonal component (S_t)



Irregular component (I_t)



Decomposition of time series

Additive decomposition

$$X_t = TC_t + S_t + I_t$$

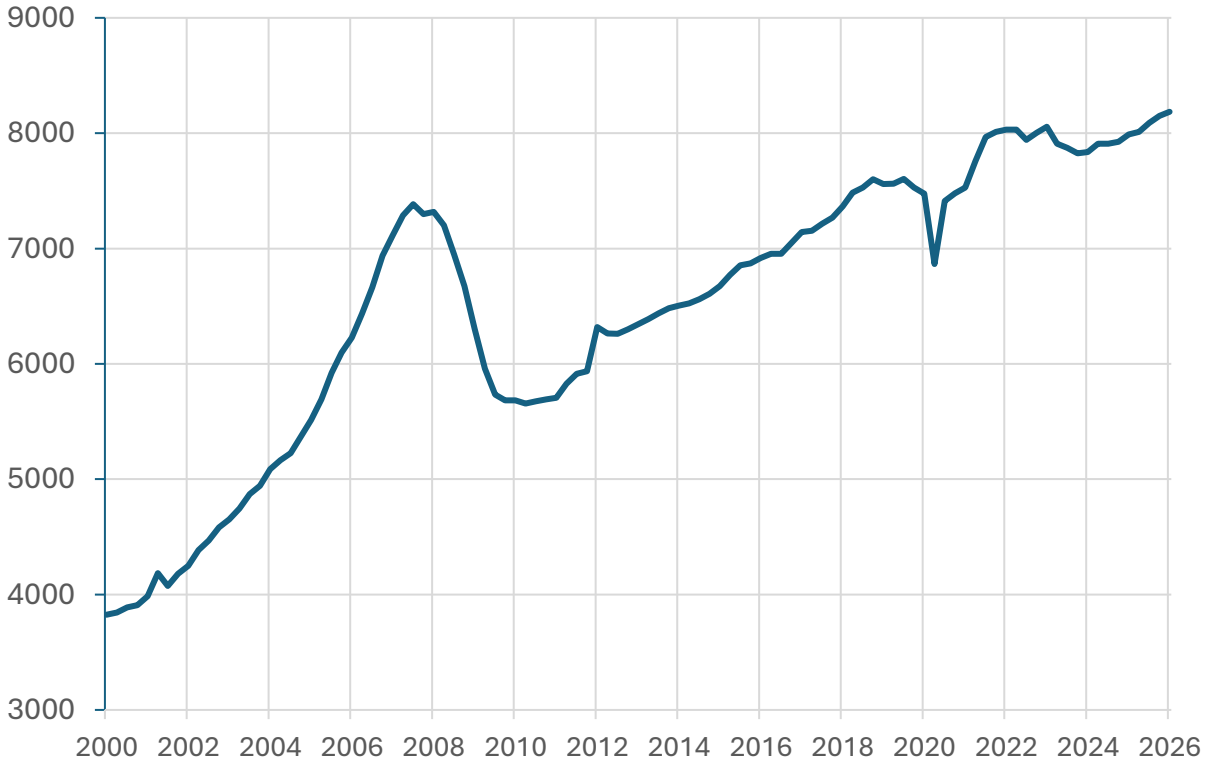
In an additive decomposition the magnitude of seasonal effects does not change as the level of the trend-cycle changes. Any series with zero or negative values are additive.

Multiplicative decomposition

$$X_t = TC_t * S_t * I_t$$

As the trend of the series increases, the magnitude of the seasonal fluctuations also increases. Most economic time series exhibit a multiplicative relation between the components.

The **seasonally adjusted series** is a combination of the trend-cycle and the irregular component.



JDemeta+



JDemetra+ is an open source software for seasonal adjustment. JDemetra+ is developed by the National Bank of Belgium in cooperation with Deutsche Bundesbank and Eurostat

JDemetra+ offers the possibility to perform seasonal adjustment by use of internationally recommended methods, TRAMO/SEATS and X-13-ARIMA-SEATS, and is officially recommended to members of the ESS and the European System of Central Banks for seasonal and calendar adjustment of official statistics.

RJDemetra is an R interface to JDemetra+.

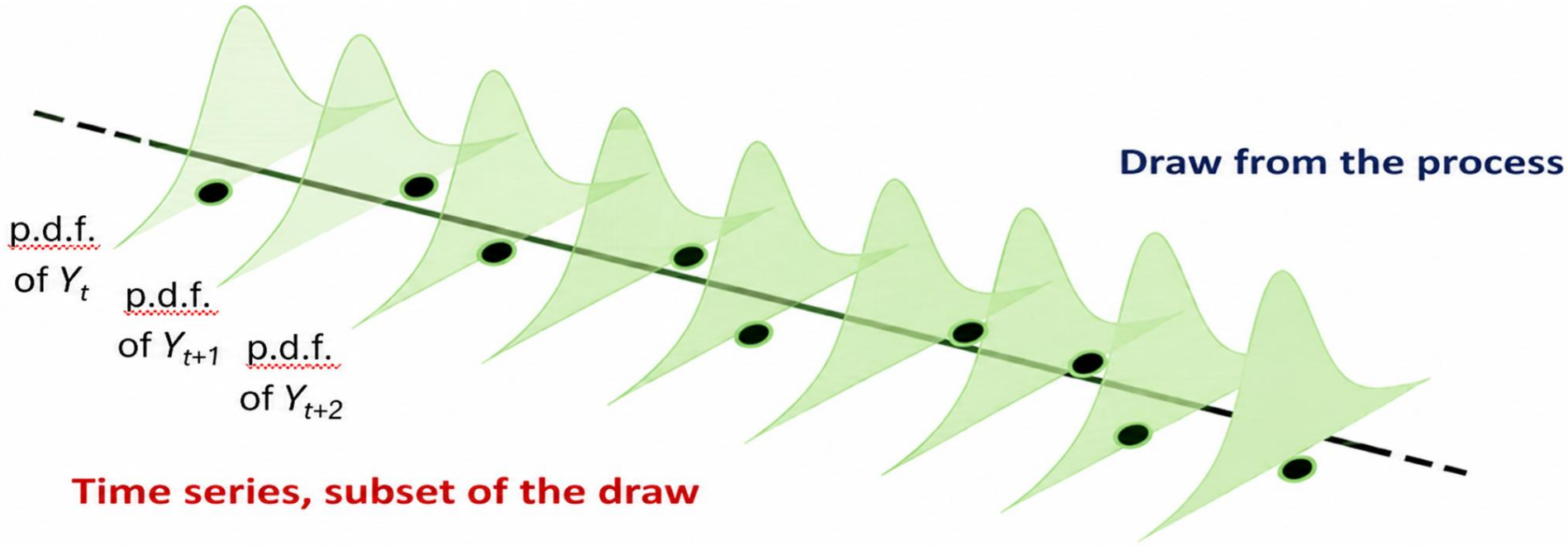
RJDemetra relies on the rJava package and Java SE version 8 or higher is required.

```
# RJDemetra package
install.packages("RJDemetra")
library(RJDemetra)
```

Stochastic process and time series

Time series is a series of values of some random variable, obtained at successive periods.

A sequence of random variables indexed by time is called a **stochastic process**.



2. Stationarity

Stationarity

Stationary stochastic processes:

- **Strict stationarity** – the whole probability distribution is unchanged over time.
- Covariance (**weakly**) **stationary**:
  - The series fluctuates around a constant mean, has a finite constant variance

$$E(Y_t) = E(Y_{t+j}) = \mu \qquad Var(Y_t) = Var(Y_{t+j}) = \sigma^2$$

  - The covariance between two observations depends only on the distance *j* between them, not on where in the sample they sit.

$$Cov(Y_t, Y_{t+j}) = Cov(Y_s, Y_{s+j}) = \gamma_j$$

Why we insist on stationarity

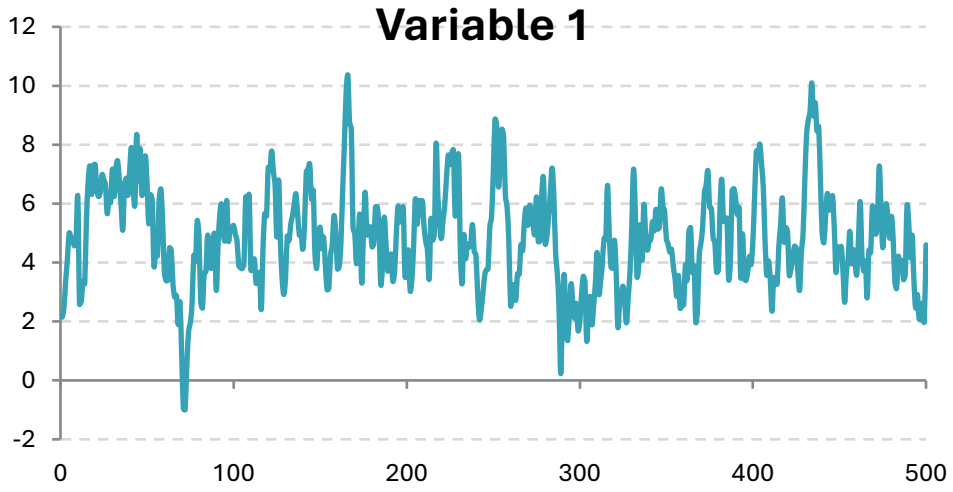
If the series is non-stationary, the standard results of classical regression analysis do not apply.

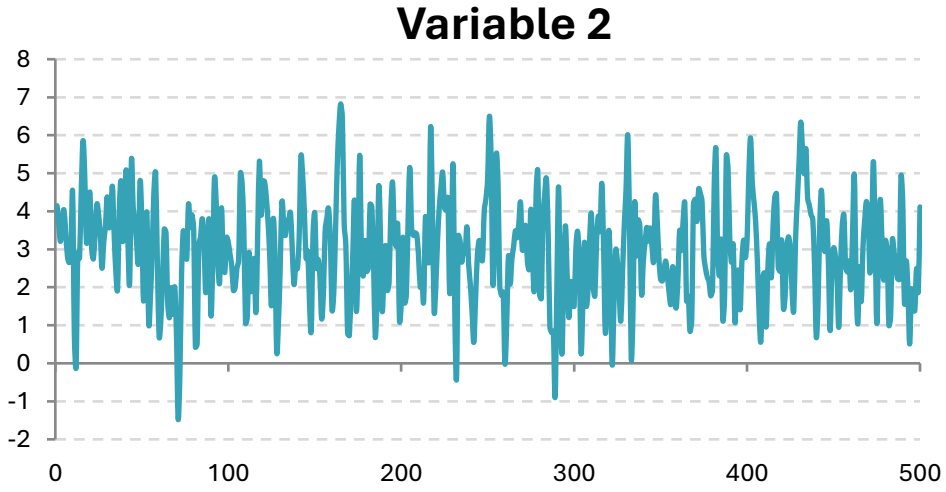
A regression between two unrelated series that both trend will report a strong relationship. This is a mechanical property of the data, not a finding. We return to it next lecture.

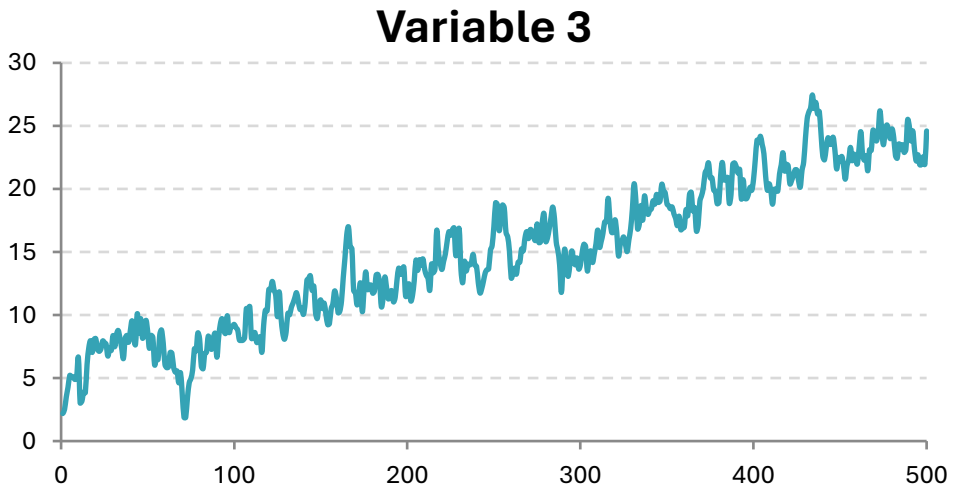
Shocks to a stationary series are necessarily temporary: their effects dissipate and the series reverts to its long-run mean. Long-horizon forecasts therefore converge on the unconditional mean.

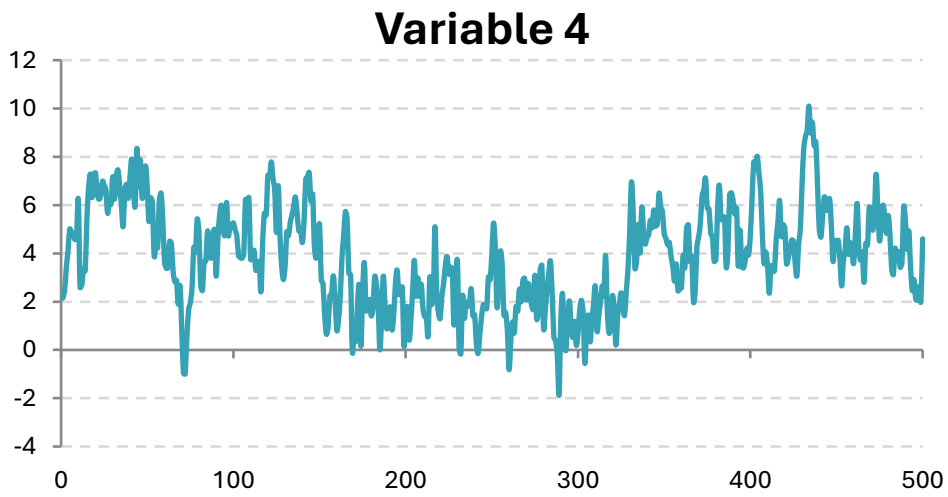
3. Identification of a time series

Visual inspection: graph and observe your data. Which series look stationary?









Three building blocks

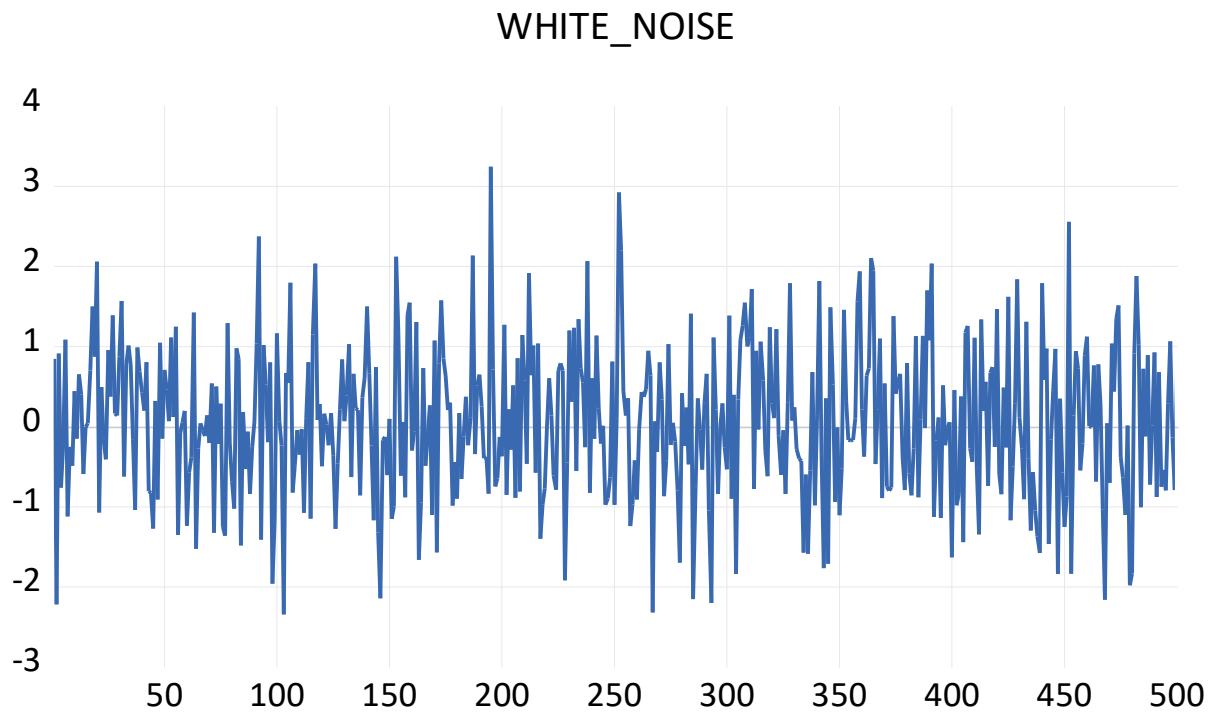
- AR(p): autoregressive model $y_t = b_0 + b_1 y_{t-1} + b_2 y_{t-2} + \cdots + b_p y_{t-p} + u_t$
- MA(q): moving average model $y_t = \mu + u_t + \phi_1 u_{t-1} + \phi_2 u_{t-2} + \cdots + \phi_q u_{t-q}$
- ARMA(p,q) $y_t = b_0 + b_1 y_{t-1} + b_2 y_{t-2} + \cdots + b_p y_{t-p} + u_t + \phi_1 u_{t-1} + \phi_2 u_{t-2} + \cdots + \phi_q u_{t-q}$

Notation and the white-noise assumption

- *p* and *q* are the orders of the process, meaning the longest AR and MA lag included.

- The disturbance u_t is assumed white noise:

$$E[u_t] = 0,$$
$$Var[u_t] = \sigma^2,$$
$$Cov[u_t, u_s] = 0, for\ t \neq s$$



Autocovariance and autocorrelation

- Relations between observations at different lags:

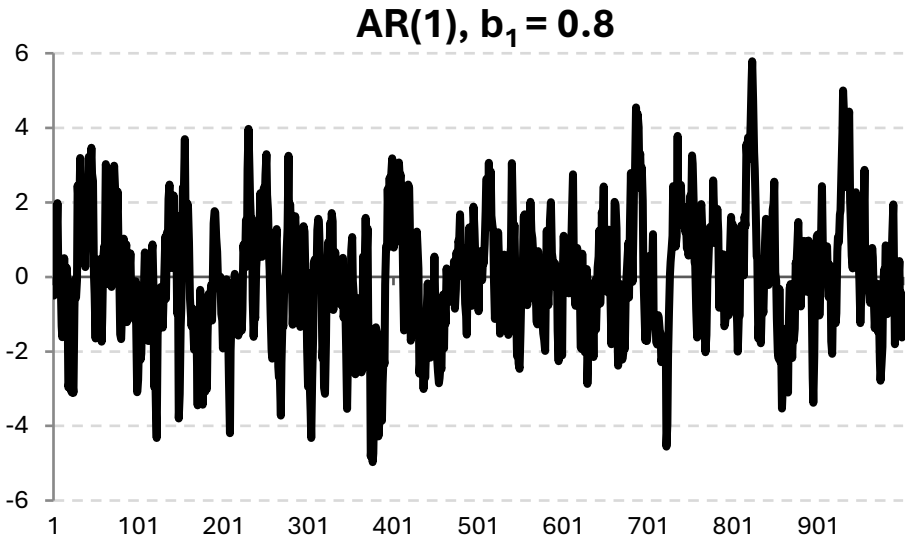
  – Autocovariance: $\gamma_j = E[(y_t - \mu)(y_{t-j} - \mu)]$

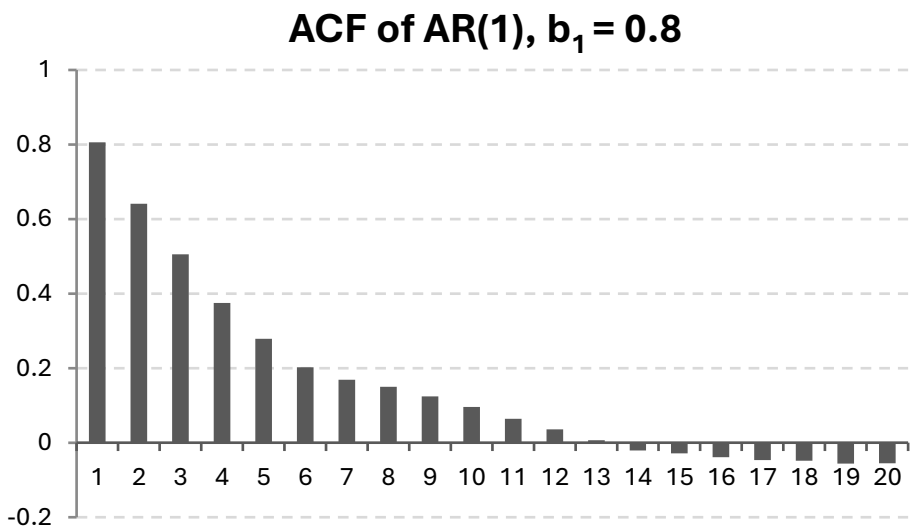
  – Autocorrelation: $\rho_j = \frac{\gamma_j}{\gamma_0}$
  (scaling the autocovariance by the variance γ_0)

  – Plotting ρ_j against *j* gives the autocorrelation function (ACF), or correlogram.

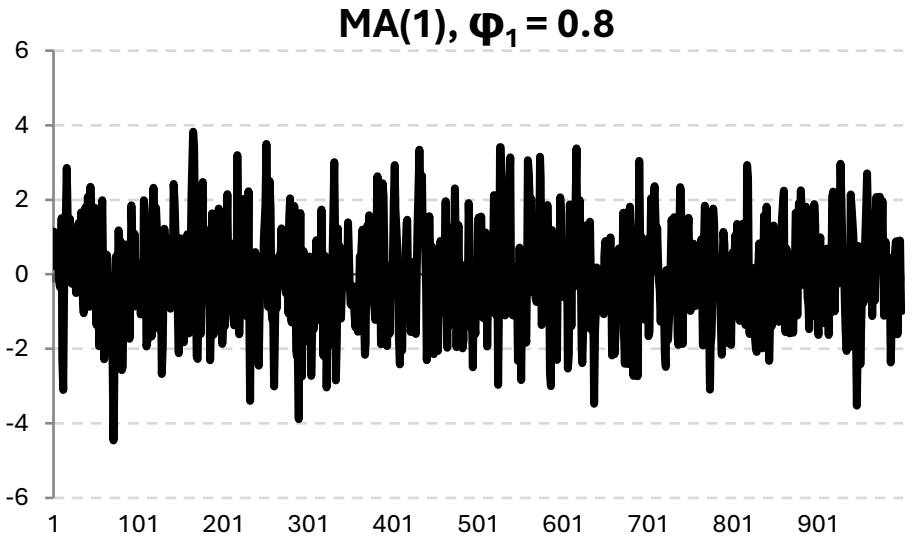
Different patterns in ACFs

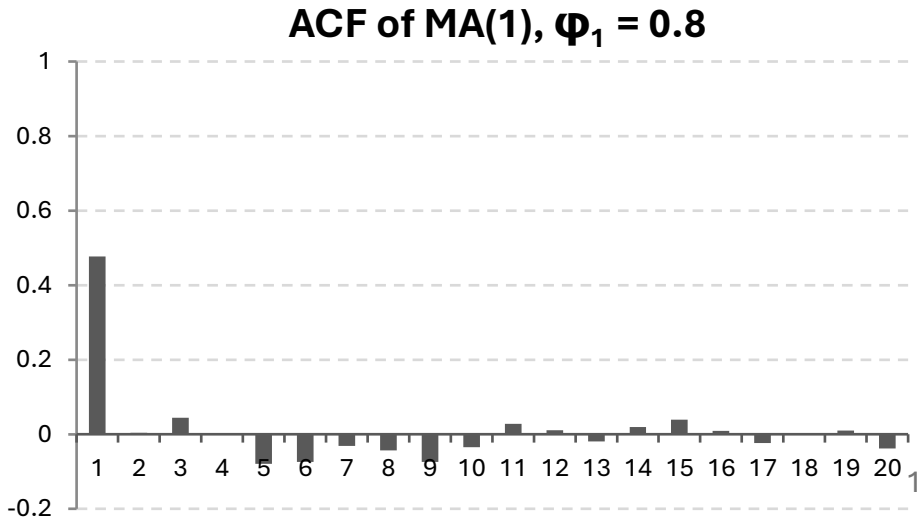
AR(1),
$b_1 = 0.8$





MA(1),
$\phi_1 = 0.8$





Partial autocorrelation

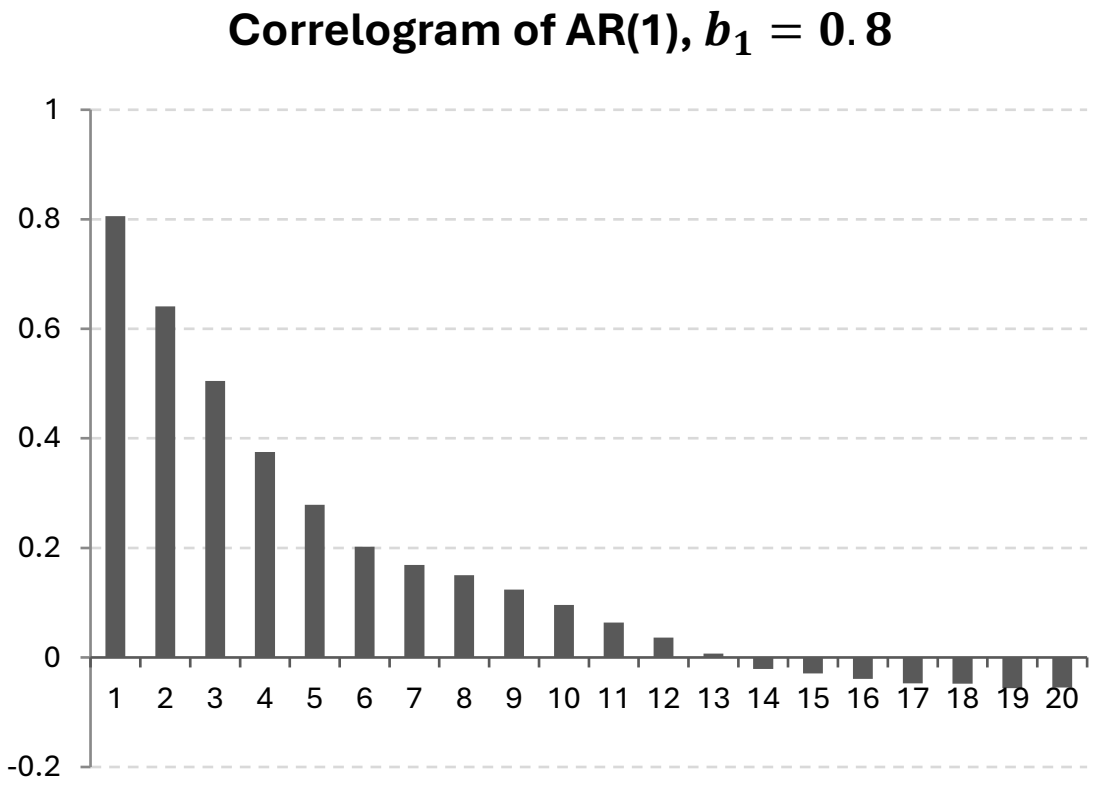
- The partial autcorrelation at lag *p* (PAC_p) is the coefficient $\hat{b}_p$ from a linear regression of y_t on all its lags up to *p*:

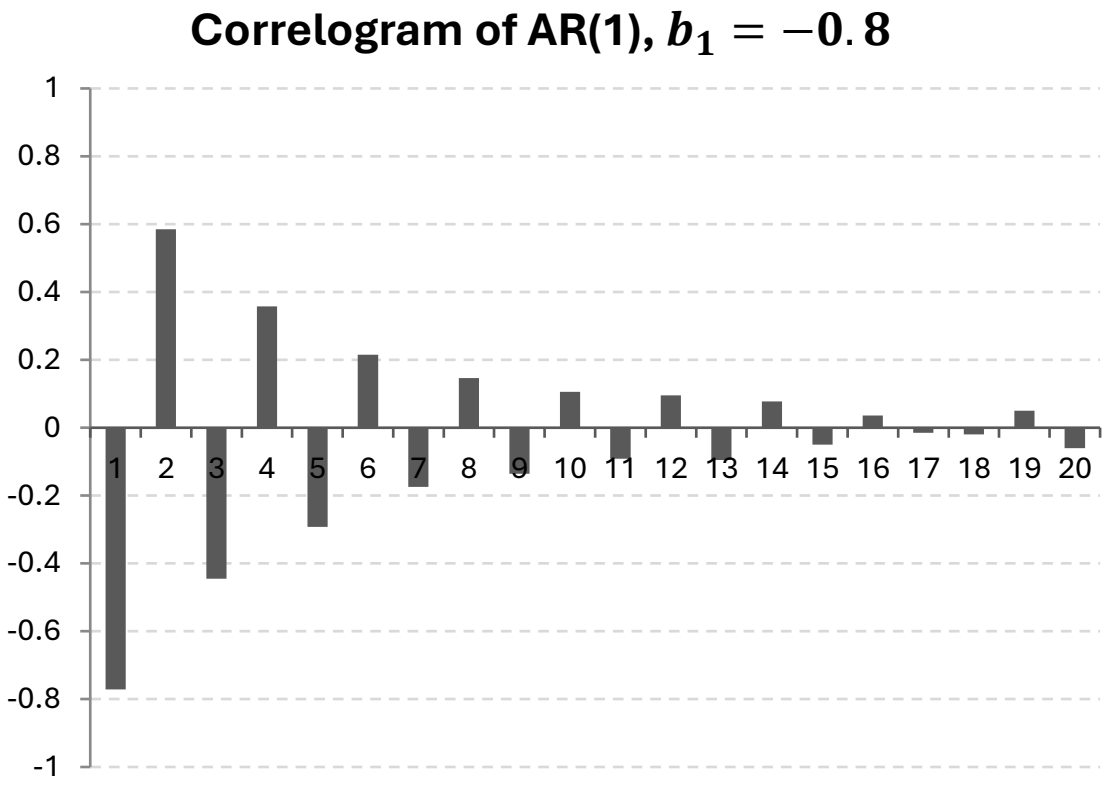
$$y_t = \hat{b}_0 + \hat{b}_1 y_{t-1} + \hat{b}_2 y_{t-2} + \cdots + \hat{b}_p y_{t-p} + e_t$$

- Measures the relationship between y_t and y_{t-p} after controlling for intervening lag.

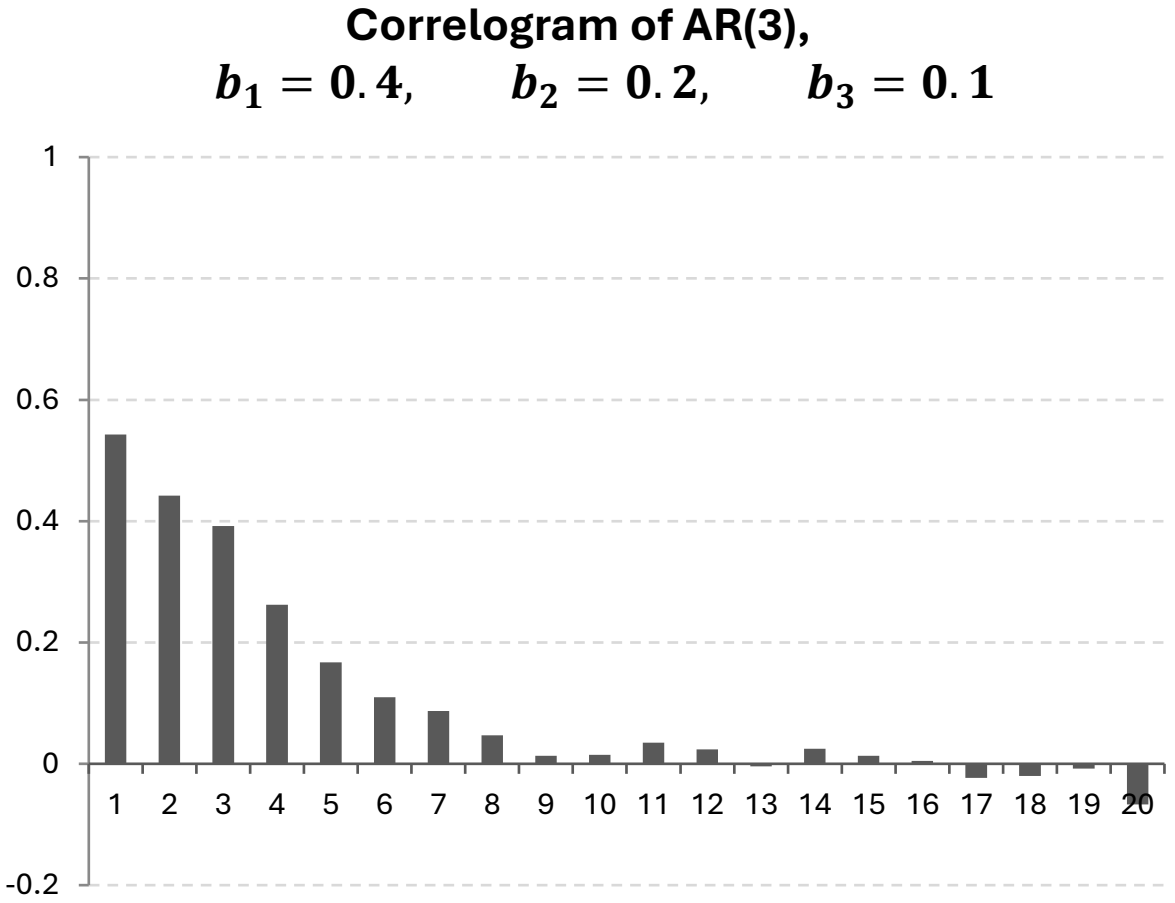
- For example, for *p* = 4, regress y_t on y_{t-1}, y_{t-2}, y_{t-3}, y_{t-4}.

The patterns. Geometric decay of ACF in AR(1), oscillating if b_1<0

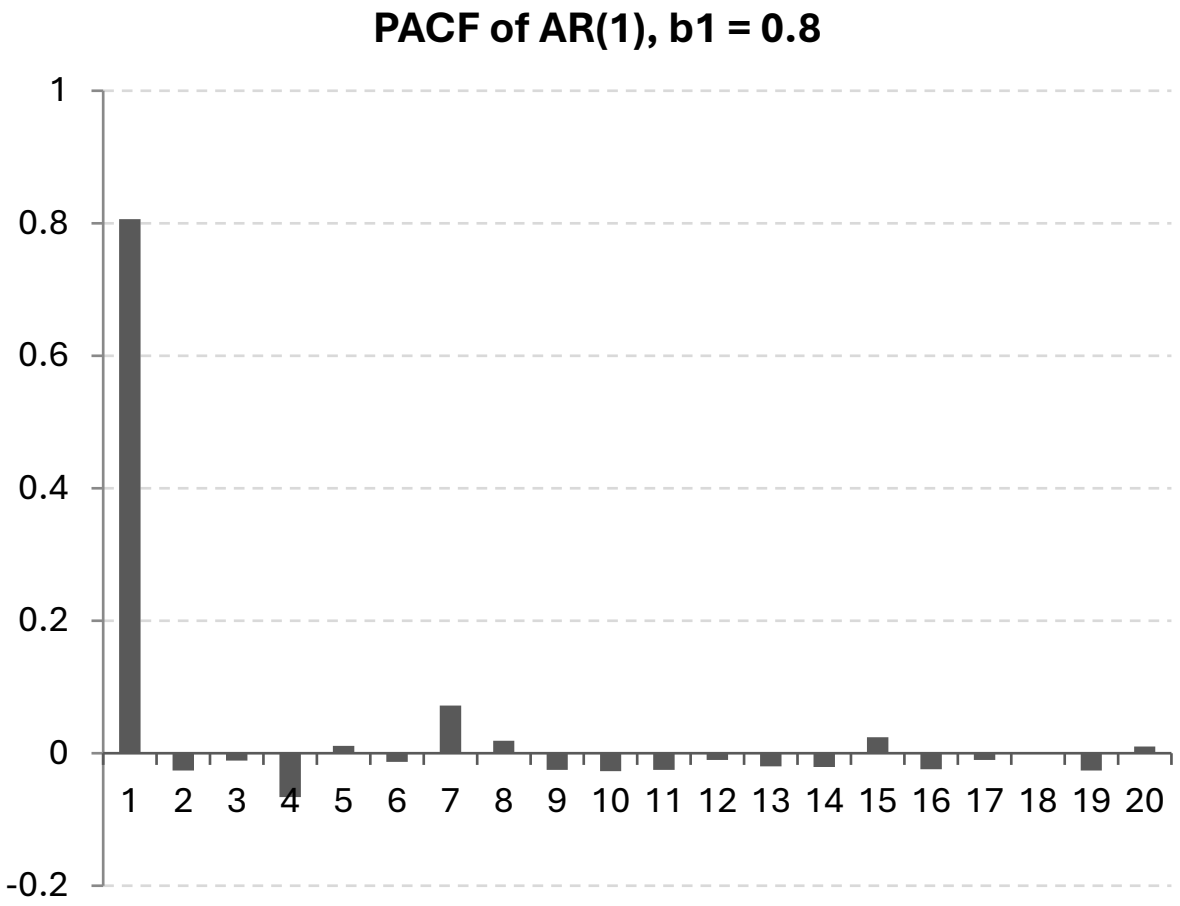


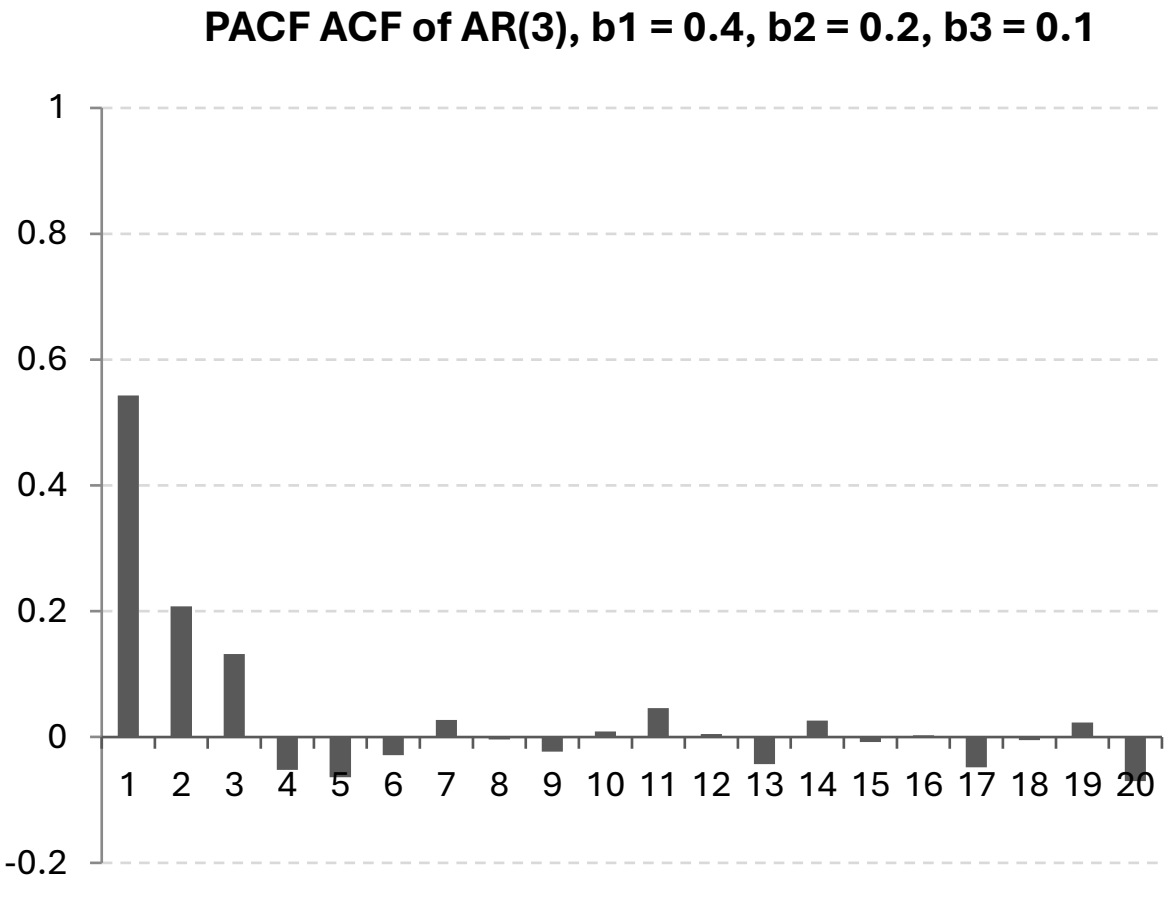


The patterns. Gradual decay of ACF in AR(p) toward zero



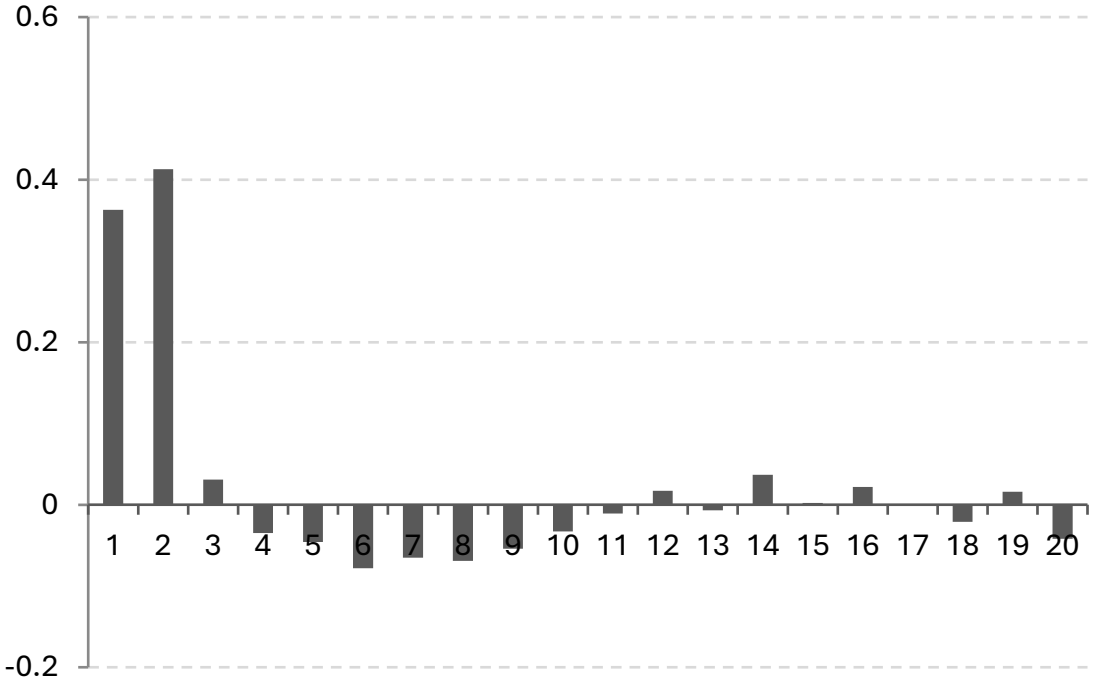
The patterns. Abrupt drop-off in PACF in AR(*p*) after lag *p*



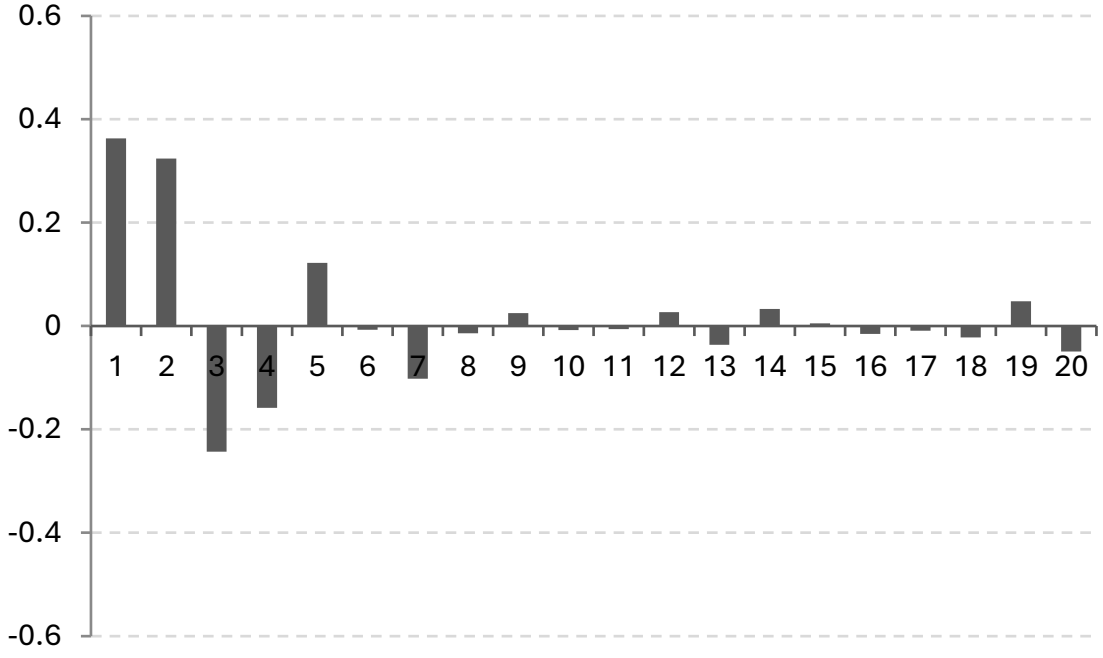


The patterns. Abrupt drop-off in ACF after q in MA(q), gradual decay toward zero in PACF

Correlogram of MA(2),
$\phi_1 = 0.3, \quad \phi_2 = 0.6$



Partial autocorrelation of MA(2),
$\phi_1 = 0.3, \quad \phi_2 = 0.6$



Summary of the patterns

- Read the two functions together (ACF and PACF). Neither identifies a process on its own.
- An ACF with no visible tendency to decay is a symptom of non-stationarity.
- Mixed ARMA processes are genuinely hard to identify, because both functions decay. When both decay gradually, estimate several candidates and compare them.

Process	ACF	PACF
AR(p)	Decays gradually toward zero (oscillating if the coefficient is negative)	Cuts off abruptly after lag p
MA(q)	Cuts off abruptly after lag q	Decays gradually toward zero
ARMA(p, q)	Decays gradually	Decays gradually

4. Estimation and diagnostic of univariate time series models

From candidates to a chosen model

- Draw up a short list of plausible specifications from the correlograms. Two or three, not ten.
- Estimate each over an identical sample.
- Judge each candidate on five things:
  - individual significance of the coefficients, and of the longest lag in particular;
  - residuals indistinguishable from white noise;
  - information criteria;
  - accuracy on data held back from estimation.

Fit against parsimony

- Information criteria formalise the trade-off between fit and parsimony:
  - Akaike Information Criterion (AIC)

$$AIC = T * \ln(SSR) + (p + q + 1) * 2$$

  - Schwarz Bayesian Criterion (SBC)

$$SBC = T * \ln(SSR) + (p + q + 1) * \ln(T)$$

- ln(*T*) exceeds 2 for any *T* above 7, so SBC penalises more heavily and selects smaller models. The gap widens with the sample.

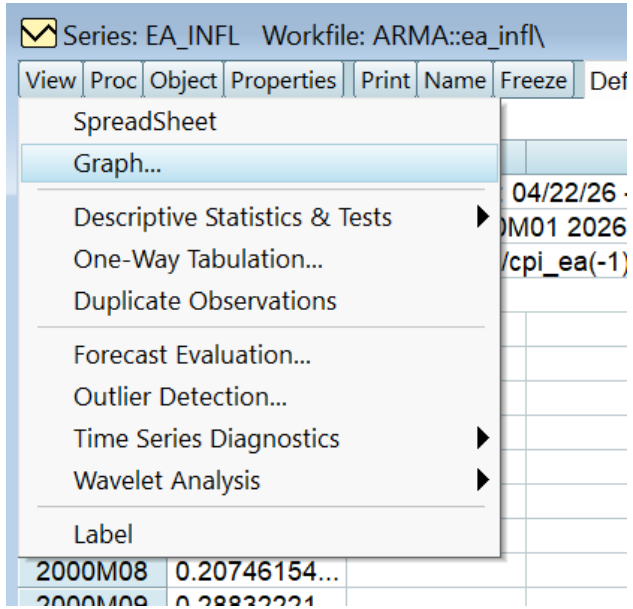
Checking the residuals

- The target is residuals with no serial correlation left. Autocorrelation in the residuals says the lag structure is misspecified, and it also invalidates the standard errors you were about to report.
- Standardised residuals (u_t/σ) under normality roughly 95% should fall inside [−2, +2].
- First look: the residual correlogram and the significance of the autocorrelations.
- Formal test: Ljung–Box Q: joint significance test up to lag s:

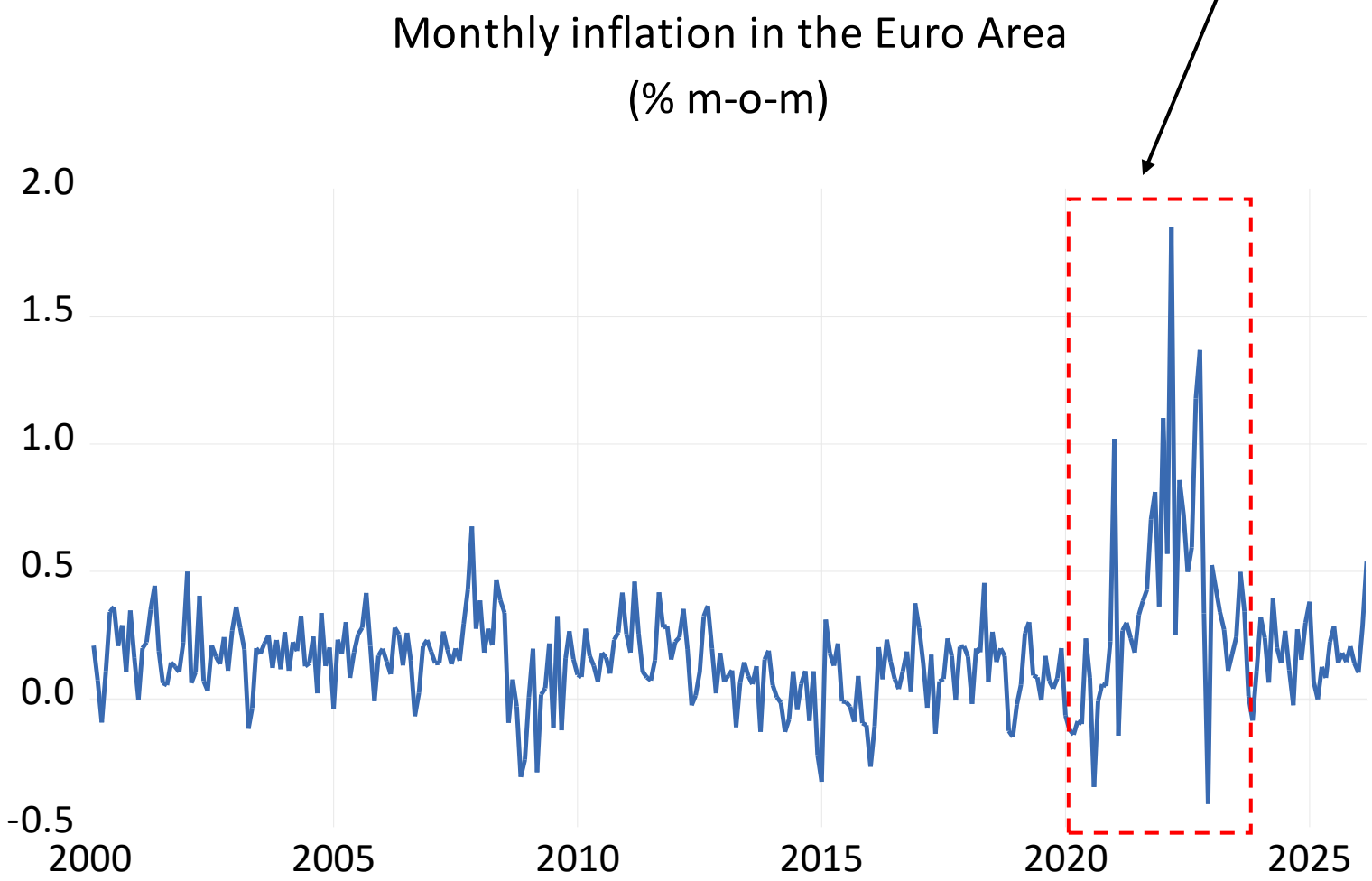
$$Q = T(T+2)\sum_{k=1}^{s} r_k^2\,/(T-k) \sim \chi^2_{s-p-q}$$

$$\mathrm{H}_0\text{: all } r_k = 0$$
$$\mathrm{H}_1\text{: at least one } r_k \neq 0$$

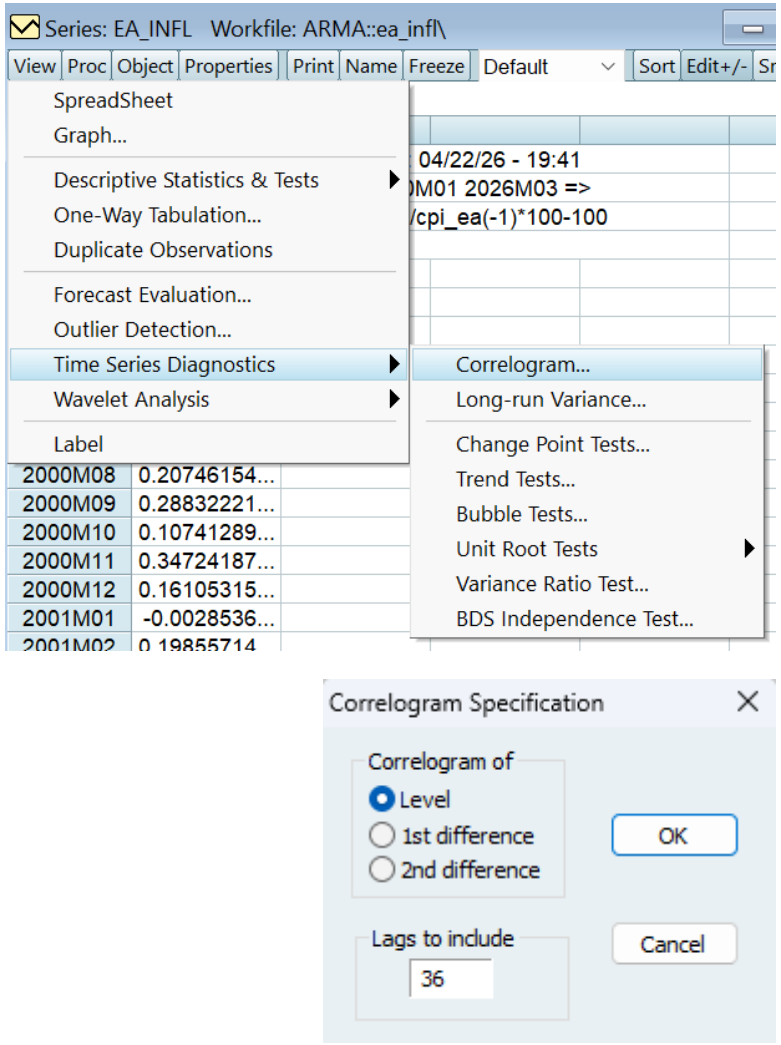
EA inflation. Identification (1)



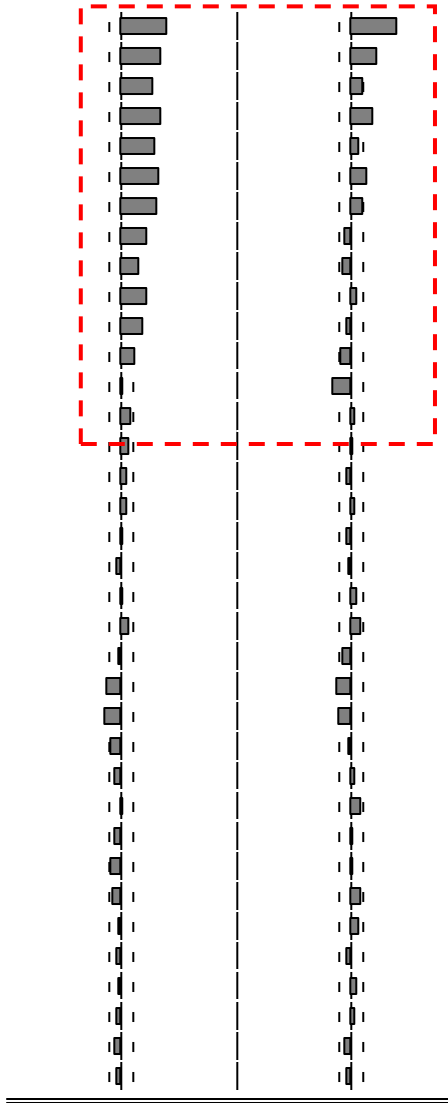
Increase in the mean and the variance – indicator of possible non-stationarity *(but unit-root tests reject the presence of a unit root)*



EA inflation. Identification (2)



Date: 04/22/26 Time: 19:05
Sample (adjusted): 2000M02 2026M03
Included observations: 314 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.417	0.417	55.237	0.000
		2	0.361	0.226	96.684	0.000
		3	0.294	0.106	124.28	0.000
		4	0.359	0.204	165.60	0.000
		5	0.300	0.072	194.45	0.000
		6	0.352	0.153	234.42	0.000
		7	0.331	0.101	269.80	0.000
		8	0.238	-0.050	288.22	0.000
		9	0.170	-0.070	297.61	0.000
		10	0.230	0.055	314.90	0.000
		11	0.190	-0.025	326.76	0.000
		12	0.125	-0.085	331.88	0.000
		13	0.019	-0.164	332.00	0.000
		14	0.095	0.026	334.99	0.000
		15	0.072	0.016	336.73	0.000
		16	0.046	-0.033	337.44	0.000
		17	0.058	0.033	338.58	0.000
		18	0.008	-0.038	338.60	0.000
		19	-0.031	-0.011	338.93	0.000
		20	-0.001	0.054	338.93	0.000
		21	0.074	0.094	340.80	0.000
		22	-0.019	-0.073	340.92	0.000
		23	-0.121	-0.138	345.88	0.000
		24	-0.148	-0.114	353.34	0.000
		25	-0.094	-0.013	356.40	0.000
		26	-0.062	0.026	357.74	0.000
		27	-0.005	0.083	357.75	0.000
		28	-0.050	-0.006	358.63	0.000
		29	-0.088	-0.000	361.34	0.000
		30	-0.074	0.093	363.26	0.000
		31	-0.021	0.069	363.42	0.000
		32	-0.042	-0.029	364.05	0.000
		33	-0.012	0.043	364.10	0.000
		34	-0.027	0.029	364.35	0.000
		35	-0.050	-0.063	365.23	0.000
		36	-0.027	-0.031	365.49	0.000

ACF:
- decays gradually
- no sharp cut-off

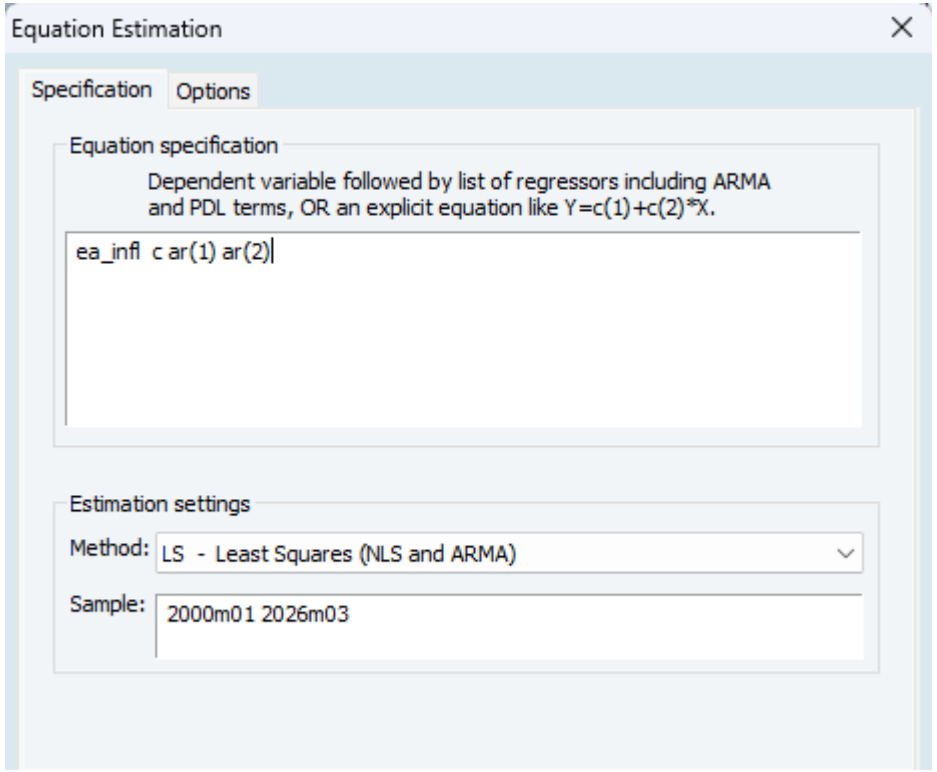
AR-type behavior

PACF:
- decays gradually
- no clear cut-off

Possible candidates:
- AR(2)
- AR(4)
- ARMA(1,1)

EA inflation. Estimation (1)

AR(2) estimation



AR(1) and AR(2) are significant at 1% level

Dependent Variable: EA_INFL
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/22/26 Time: 19:24
Sample: 2000M02 2026M03
Included observations: 314
Convergence achieved after 57 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.179558	0.031413	5.716120	0.0000
AR(1)	0.324240	0.030154	10.75284	0.0000
AR(2)	0.226554	0.027878	8.126719	0.0000
SIGMASQ	0.041700	0.001588	26.25259	0.0000

R-squared	0.218090	Mean dependent var	0.178045
Adjusted R-squared	0.210523	S.D. dependent var	0.231303
S.E. of regression	0.205519	Akaike info criterion	-0.312952
Sum squared resid	13.09376	Schwarz criterion	-0.265189
Log likelihood	53.13339	Hannan-Quinn criter.	-0.293866
F-statistic	28.82162	Durbin-Watson stat	2.035912
Prob(F-statistic)	0.000000		

Inverted AR Roots	.66	-.34

EA inflation. Estimation (2)

ARMA(1,1) has lower AIC and SBC – preferred model.

AR(2)

Dependent Variable: EA_INFL
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/22/26 Time: 19:24
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Adjusted R-squared	0.210523	S.D. dependent var	0.231303
S.E. of regression	0.205519	Akaike info criterion	-0.312952
Sum squared resid	13.09376	Schwarz criterion	-0.265189
Log likelihood	53.13339	Hannan-Quinn criter.	-0.293866
F-statistic	28.82162	Durbin-Watson stat	2.035912
Prob(F-statistic)	0.000000		

Inverted AR Roots	.66	-.34

AR(4) model

Dependent Variable: EA_INFL
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/22/26 Time: 19:15
Sample: 2000M02 2026M03
Included observations: 314
Convergence achieved after 65 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.179567	0.042637	4.211542	0.0000
AR(1)	0.278022	0.032792	8.478268	0.0000
AR(2)	0.152312	0.043002	3.541951	0.0005
AR(3)	0.044634	0.050130	0.890363	0.3740
AR(4)	0.203090	0.055285	3.673520	0.0003
SIGMASQ	0.039523	0.001820	21.71407	0.0000

R-squared	0.258902	Mean dependent var	0.178045
Adjusted R-squared	0.246871	S.D. dependent var	0.231303
S.E. of regression	0.200732	Akaike info criterion	-0.353188
Sum squared resid	12.41032	Schwarz criterion	-0.281543
Log likelihood	61.45046	Hannan-Quinn criter.	-0.324560
F-statistic	21.51993	Durbin-Watson stat	2.017289
Prob(F-statistic)	0.000000		

Inverted AR Roots	.85	.03-.61i	.03+.61i	-.64

ARMA (1,1) model

Dependent Variable: EA_INFL
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/22/26 Time: 19:19
Sample: 2000M02 2026M03
Included observations: 314
Convergence achieved after 73 iterations
Coefficient covariance computed using outer product of gradients

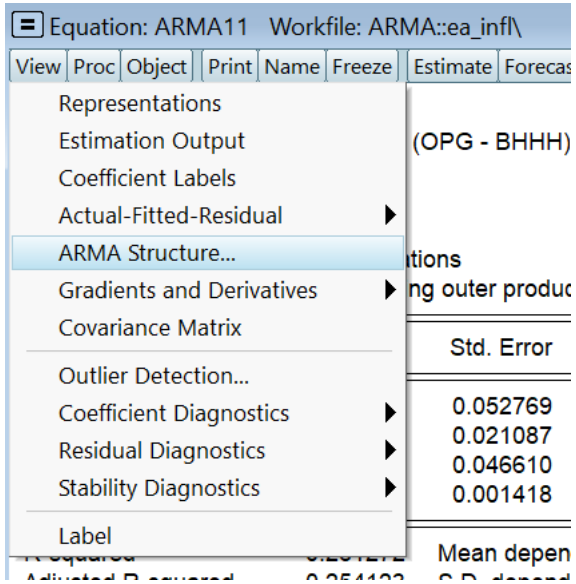
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.180439	0.052769	3.419415	0.0007
AR(1)	0.923720	0.021087	43.80515	0.0000
MA(1)	-0.699270	0.046610	-15.00246	0.0000
SIGMASQ	0.039397	0.001418	27.78007	0.0000

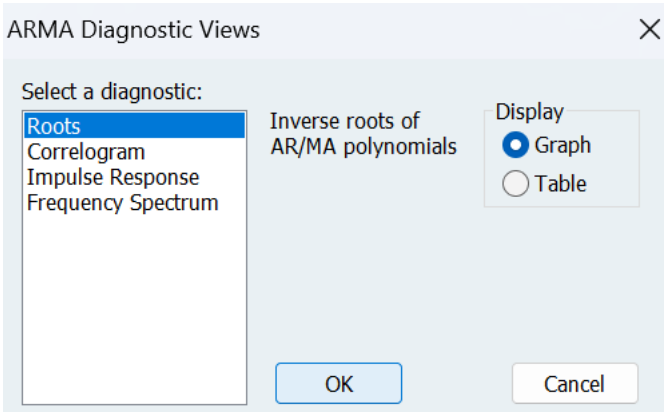
R-squared	0.261272	Mean dependent var	0.178045
Adjusted R-squared	0.254123	S.D. dependent var	0.231303
S.E. of regression	0.199763	Akaike info criterion	-0.369076
Sum squared resid	12.37064	Schwarz criterion	-0.321313
Log likelihood	61.94490	Hannan-Quinn criter.	-0.349991
F-statistic	36.54667	Durbin-Watson stat	1.949070
Prob(F-statistic)	0.000000		

Inverted AR Roots	.92
Inverted MA Roots	.70

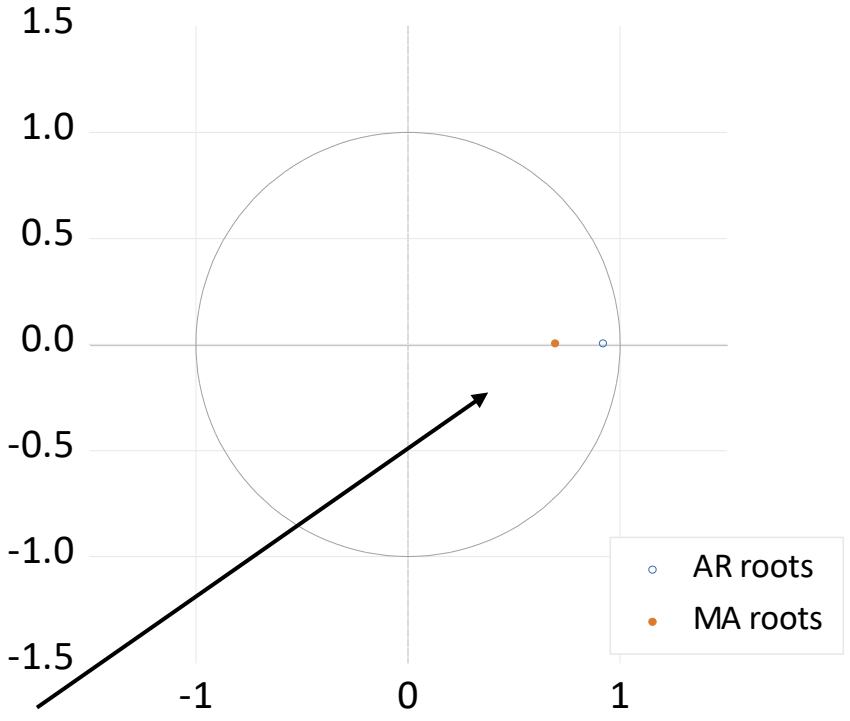
EA inflation. Diagnostic (1)

The model is stationary





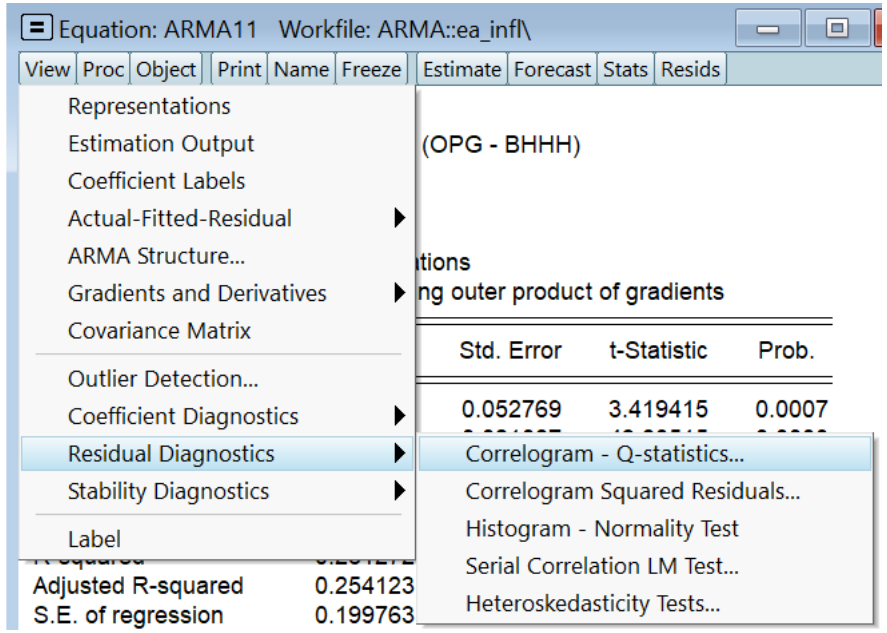
EA_INFL: Inverse Roots of AR/MA Polynomial(s)



All inverted roots must be within the unit circle

EA inflation. Diagnostic (2)

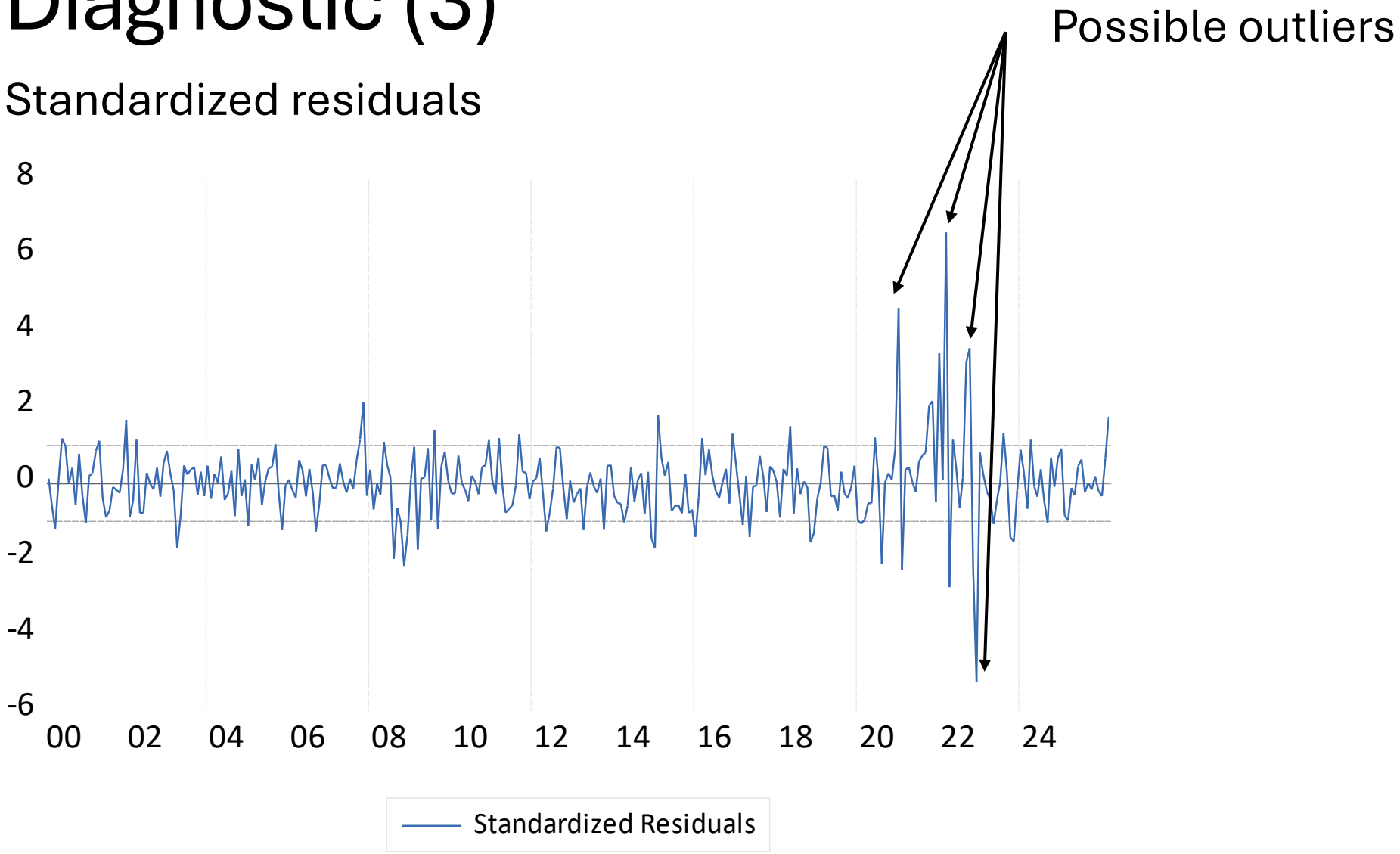
Residual autocorrelation !



Date: 04/22/26 Time: 19:56
Sample (adjusted): 2000M02 2026M03
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.020	0.020	0.1327	
		2	-0.023	-0.023	0.3004	
		3	-0.094	-0.094	3.1461	0.076
		4	0.055	0.059	4.1201	0.127
		5	-0.014	-0.021	4.1861	0.242
		6	0.111	0.107	8.1414	0.087
		7	0.114	0.122	12.378	0.030
		8	0.002	-0.005	12.379	0.054
		9	-0.069	-0.043	13.945	0.052
		10	0.079	0.092	15.977	0.043
		11	0.054	0.039	16.930	0.050
		12	-0.012	-0.029	16.975	0.075
		13	-0.154	-0.161	24.765	0.010
		14	0.010	-0.004	24.798	0.016
		15	0.001	0.002	24.798	0.025
		16	-0.016	-0.048	24.888	0.036
		17	0.031	0.020	25.216	0.047
		18	-0.028	-0.048	25.474	0.062
		19	-0.075	-0.034	27.370	0.053
		20	-0.005	0.045	27.378	0.072
		21	0.146	0.136	34.579	0.016
		22	0.011	-0.006	34.622	0.022
		23	-0.134	-0.106	40.726	0.006
		24	-0.161	-0.137	49.562	0.001
		25	-0.067	-0.075	51.085	0.001
		26	-0.016	-0.044	51.176	0.001
		27	0.079	0.032	53.313	0.001
		28	-0.001	-0.047	53.314	0.001
		29	-0.069	-0.056	54.990	0.001
		30	-0.052	0.050	55.926	0.001
		31	0.033	0.064	56.316	0.002

EA inflation. Diagnostic (3)



EA inflation. Re-estimation

Model's fit improves. But coefficients don't change much.

ARMA (1,1) model

Dependent Variable: EA_INFL
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/22/26 Time: 19:19
Sample: 2000M02 2026M03
Included observations: 314
Convergence achieved after 73 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.180439	0.052769	3.419415	0.0007
AR(1)	0.923720	0.021087	43.80515	0.0000
MA(1)	-0.699270	0.046610	-15.00246	0.0000
SIGMASQ	0.039397	0.001418	27.78007	0.0000

R-squared	0.261272	Mean dependent var	0.178045
Adjusted R-squared	0.254123	S.D. dependent var	0.231303
S.E. of regression	0.199763	Akaike info criterion	-0.369076
Sum squared resid	12.37064	Schwarz criterion	-0.321313
Log likelihood	61.94490	Hannan-Quinn criter.	-0.349991
F-statistic	36.54667	Durbin-Watson stat	1.949070
Prob(F-statistic)	0.000000		

Inverted AR Roots	.92
Inverted MA Roots	.70

Adjusted ARMA (1,1) model

Dependent Variable: EA_INFL
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/22/26 Time: 20:00
Sample: 2000M02 2026M03
Included observations: 314
Convergence achieved after 40 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.167928	0.033357	5.034226	0.0000
D2021M1	0.905660	0.338732	2.673675	0.0079
D2022M3	1.292574	0.284618	4.541432	0.0000
D2022M9	0.900945	0.102167	8.818322	0.0000
AR(1)	0.894556	0.033859	26.42005	0.0000
MA(1)	-0.635165	0.058917	-10.78064	0.0000
SIGMASQ	0.026181	0.001588	16.48710	0.0000

R-squared	0.509088	Mean dependent var	0.178045
Adjusted R-squared	0.499493	S.D. dependent var	0.231303
S.E. of regression	0.163639	Akaike info criterion	-0.758843
Sum squared resid	8.220747	Schwarz criterion	-0.675258
Log likelihood	126.1384	Hannan-Quinn criter.	-0.725444
F-statistic	53.06103	Durbin-Watson stat	1.880570
Prob(F-statistic)	0.000000		

Inverted AR Roots	.89
Inverted MA Roots	.64

Date: 04/22/26 Time: 20:03
Sample (adjusted): 2000M02 2026M03
Q-statistic probabilities adjusted for 2 ARMA terms



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	0.052	0.052	0.8620	
		2	-0.064	-0.067	2.1585	
		3	-0.048	-0.041	2.8806	0.090
		4	-0.024	-0.024	3.0663	0.216
		5	0.011	0.008	3.1059	0.376
		6	0.112	0.107	7.1145	0.130
		7	0.024	0.013	7.3065	0.199
		8	0.027	0.040	7.5384	0.274
		9	0.050	0.059	8.3429	0.303
		10	0.049	0.055	9.1147	0.333
		11	0.002	0.007	9.1166	0.427
		12	-0.086	-0.089	11.573	0.315
		13	-0.087	-0.079	14.043	0.231
		14	-0.048	-0.061	14.804	0.252
		15	0.088	0.063	17.371	0.183
		16	-0.036	-0.077	17.800	0.216
		17	-0.008	-0.007	17.824	0.272
		18	-0.047	-0.036	18.563	0.292
		19	0.014	0.036	18.625	0.351
		20	0.012	0.022	18.672	0.412
		21	0.039	0.041	19.187	0.445
		22	0.004	0.035	19.191	0.509
		23	-0.025	-0.005	19.411	0.559
		24	-0.176	-0.174	29.959	0.119
		25	-0.044	-0.054	30.614	0.133
		26	0.054	0.021	31.634	0.136
		27	0.009	-0.016	31.663	0.168
		28	-0.042	-0.058	32.275	0.184
		29	-0.016	-0.015	32.359	0.219
		30	0.002	0.018	32.360	0.260

Summary

In the **identification** stage, the researcher visually examines the time plot of the series autocorrelation function, and partial correlation function. The stationary time series has the constant mean, variance and covariance. Once we have achieved stationarity, the next step is to identify the *p* and *q* orders of the ARMA model by using the ACF and PACF.

In the **estimation** stage, each of the tentative models is estimated and the various coefficients are examined. In this second stage, the estimated models are compared using the Akaike information criterion (AIC) and the Schwarz Bayesian criterion (SBC). We will choose the model with the smallest AIC and SBC values. Of the two criteria, the SBC is preferable.

In the **diagnostic** checking stage, we examine the goodness of fit of the model. The standard practice at this stage is to plot the residuals and look for outliers and evidence of periods in which the model does not fit the data well. We must be careful here to avoid overfitting (the procedure of adding another coefficient in an appropriate model). The special statistic that we use here is the Ljung-Box Q-statistic, which serve to test for autocorrelations of the residuals.

Appendix.
JDemetra+ quick guide

A1. Preparation of data

Seasonal adjustment.
Step 1. Data preparation: Length of time series

Length of the original time series

- For **monthly series**, observations for at least three years (**36 months**) should be available.
- For **quarterly data**, the series should be at least four years long (**16 quarters**).

Very long time series do not necessarily lead to higher quality seasonal adjustment.

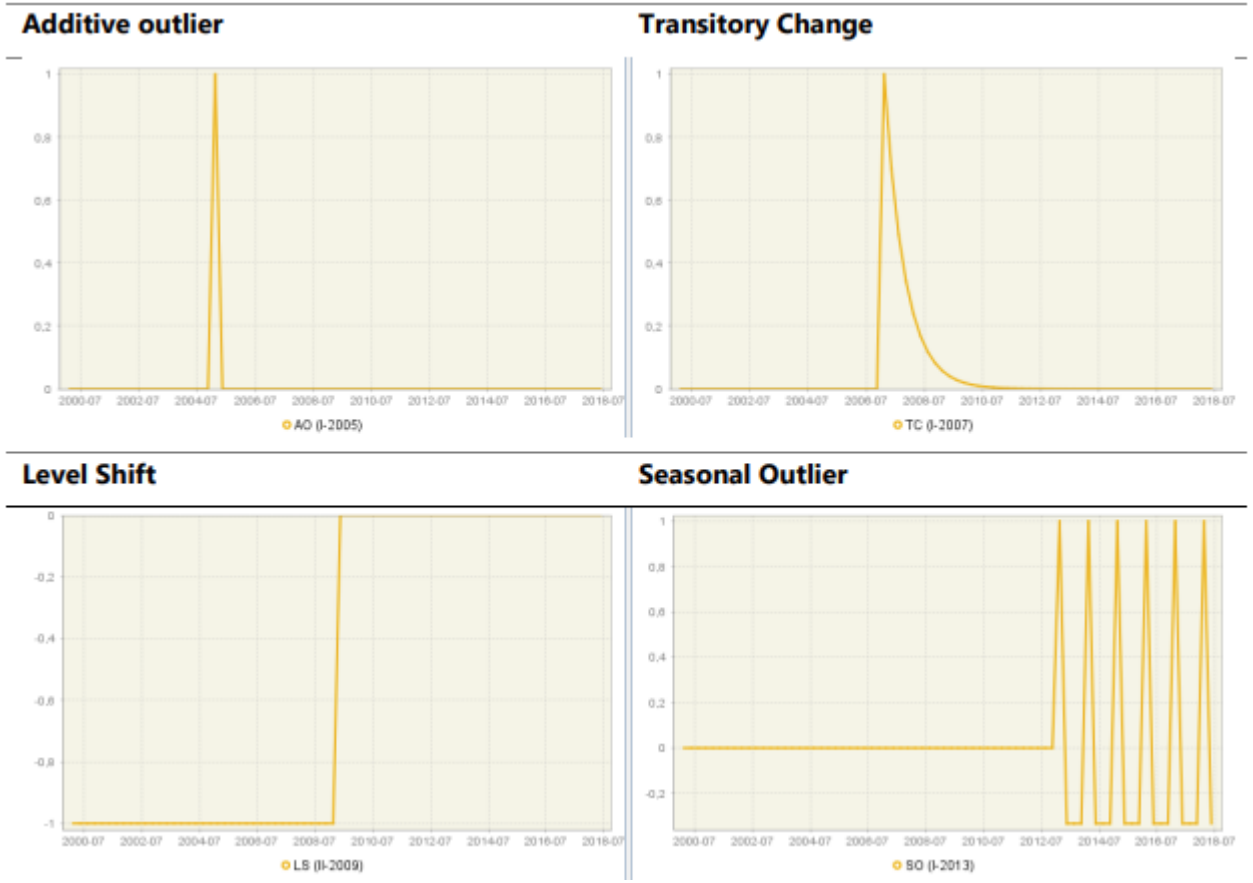
Chain-linking of the original series to avoid the risk of artificial seasonality.

Chain-linking refers to the practice where series are linked or spliced together by use of an overlap period to produce a continues time series that allows new weighting information to be introduced in the overlap period.

Seasonal adjustment.
Step 1. Data preparation: Outliers

Outliers – values that deviate from the normal or expected development by a large margin.

Many outliers in a time series may make it difficult to identify a seasonal model and tend to reduce the quality of the seasonal adjustment.



Seasonal adjustment.
Step 1. Data preparation: Criteria

To be processed in JDemetra+, the spreadsheet must meet the following criteria:

- Dates in the first column must be in a format recognized by Excel (e.g. “dd.mm.yyyy”).
  - Monthly data entries can be the first days of each month, such as “1.1.2000, 1.2.2000” (dd.mm.yyyy).
  - Quarterly data can be specified by the dates of the first day of the quarter, such as “1.1.2000, 1.4.2000, 1.7.2000” (dd.mm.yyyy). There must be three months intervals between the dates.
- The first row contains the names of the variables.
- Time series data comes in the number format.
- Top-left cell [A1] is empty.
- Missing values to be estimated are manually set to -99999. JDemetra+ uses the “Additive Outlier Approach” (Gómez, Maravall, & Pena, 1999) for the estimation of missing observations.

A2. Seasonal adjustment

Seasonal adjustment.
Step 2. Adjustment: Methods

X-13- ARIMA-SEATS

- developed by the US Census Bureau;
- a semi-parametric approach based on moving averages;
- the major disadvantage of this approach is that it uses similar types of filters for different types of time series.

TRAMO/SEATS

- developed by the Central Bank of Spain;
- parametric method based on econometric model estimation;
- the advantage of this approach is the use of data specific filters.

There is no international consensus on which of the methods in general should be the preferred one.

To start, the RSAfull specification in TRAMO/SEATS or RSA4c or RSA5c specifications of X-13 are suggested for time series with a significant calendar effect. The RSA3 specifications for both methods are a good benchmark for a time series without calendar effect.

Seasonal adjustment.
Step 2. Adjustment: TRAMO/SEATS

Summary definitions of specifications for TRAMO/SEATS

Name	Transformation	Calendar effect	Outlier detection	Modelling
RSA0	Level	None	None	Airline model with mean
RSA1	Test for log/level	None	Automatic	Airline model with mean
RSA2	Test for log/level	Test for working day and/or leap year and/or Easter	Automatic	Airline model with mean
RSA3	Test for log/level	None	Automatic	Automatic model identification
RSA4	Test for log/level	Test for working day and/or leap year and/or Easter	Automatic	Automatic model identification
RSA5	Test for log/level	Test for trading day and/or leap year and/or Easter	Automatic	Automatic model identification
RSAfull	Test for log/level	Test for trading or working day and/or leap year and/or Easter	Automatic	Automatic model identification

Seasonal adjustment.
Step 2. Adjustment: X-13-ARIMA-SEATS

Summary definitions of specifications for X-13- ARIMA-SEATS

Name	Transformation	Calendar effect	Outlier detection	Modelling
X-11	Non-applicable	Non-applicable	Non-applicable	Non-applicable
RSA0	Level	None	None	Airline model with mean / without mean
RSA1	Test for log/level	None	Automatic	Airline model with mean / without mean
RSA2c	Test for log/level	Test for working day and/or leap year and/or Easter	Automatic	Airline model with mean / without mean
RSA3	Test for log/level	None	Automatic	Automatic model identification
RSA4c	Test for log/level	Test for working day and/or leap year and/or Easter	Automatic	Automatic model identification
RSA5c	Test for log/level	Test for trading day and/or leap year and/or Easter	Automatic	Automatic model identification

A3. Seasonal adjustment in R

Seasonal adjustment.
Step 1. Data preparation: R code

In RJDemetra you need to transform your data into a time series object to make seasonal adjustment.

```r
## Seasonal Adjustment with JDemetra+

# Official JDemetra+ package
install.packages("RJDemetra")
library(RJDemetra)

# Downloading GDP data from Eurostat
gdp_eu <- get_eurostat("namq_10_gdp",
                       time_format = "date")

# Check values in the dataset
unique(gdp_eu$freq)
unique(gdp_eu$unit)
unique(gdp_eu$s_adj)
unique(gdp_eu$na_item)
unique(gdp_eu$geo)
unique(gdp_eu$TIME_PERIOD)
```

```r
# Transforming data: Leave only data on Lithuanian GDP:
# - constant 2020 prices,
# - raw data (not seasonally adjusted),
# - chain-linked volume

gdp_lt <- gdp_eu %>%
  filter(unit == "CLV20_MEUR",
         s_adj == "NSA",
         na_item == "B1GQ",
         geo == "LT") %>%
   select(TIME_PERIOD, values)

# Transforming data to the time series object
gdp_lt_ts <- ts(gdp_lt$values,
                start = c(1995, 1), frequency = 4)

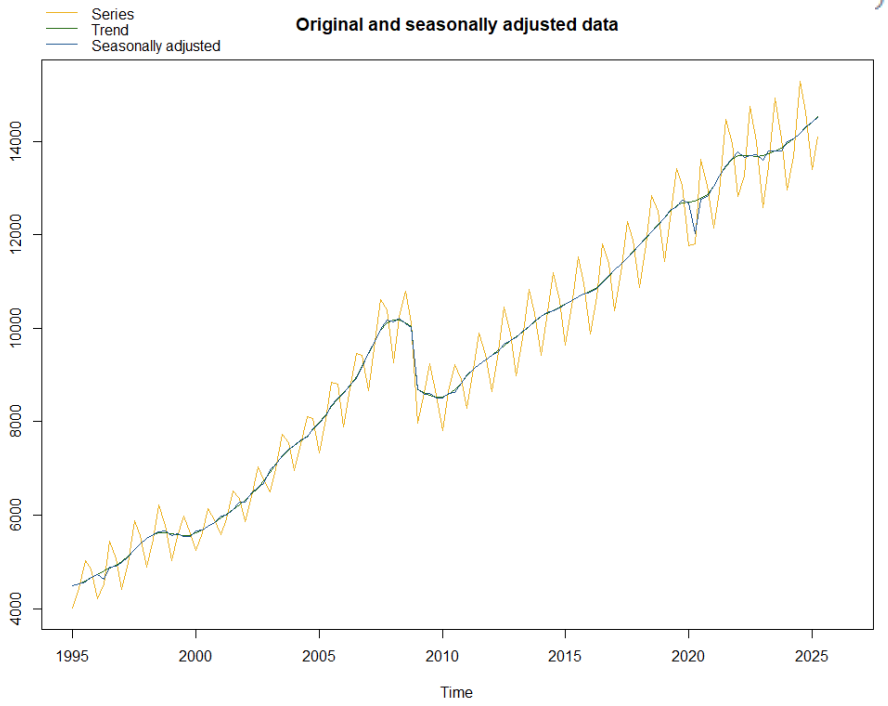
plot(gdp_lt_ts,
     main = "Original data",
     ylab = "",
     xlab = "",
     lwd = 2)
```

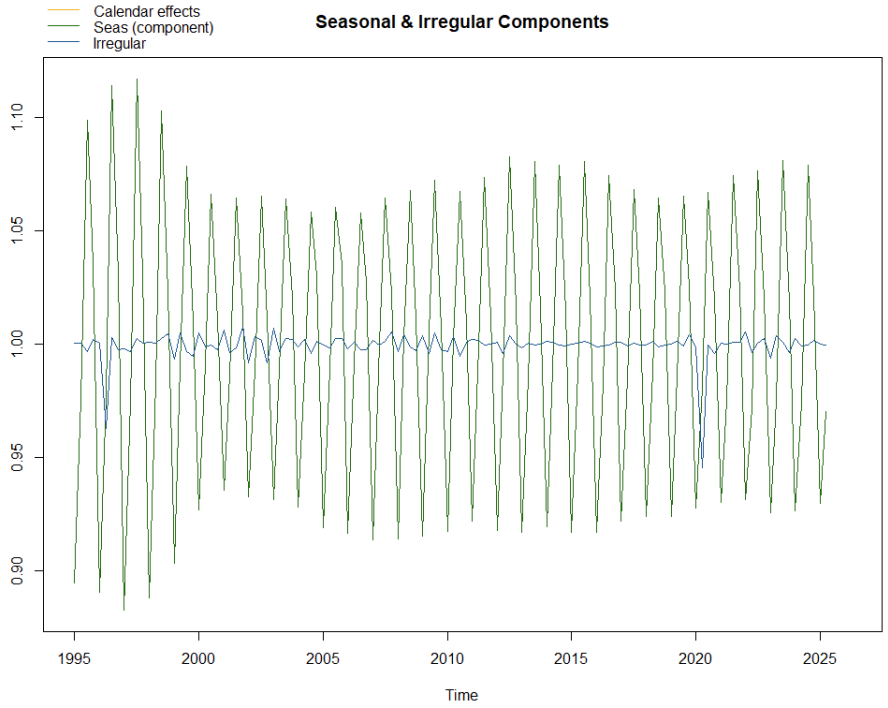
Seasonal adjustment.
Step 2. Adjustment: R code TRAMO/SEATS (1)

```r
# Seasonal Adjustment: TRAMO/SEATS
gdp_lt_sa <- tramoseats(
  gdp_lt_ts, spec = c("RSAfull"))
```

```r
# Results

plot(
  gdp_lt_sa,
  forecast = FALSE,
  type_chart = c("sa-trend", "cal-seas-irr"),
  caption = c(`sa-trend` = "Original and seasonally adjusted data", `cal-seas-irr` =
                "Seasonal & Irregular Components"),
  ask = FALSE
)
```





Seasonal adjustment.
Step 2. Adjustment: R code TRAMO/SEATS (2)

```
gdp_nsa <- gdp_lt_sa$final$series[, "y"]    # original time series
gdp_sa <- gdp_lt_sa$final$series[, "sa"]    # seasonally adjusted time series
gdp_t <- gdp_lt_sa$final$series[, "t"]      # trend-cycle components
gdp_s <- gdp_lt_sa$final$series[, "s"]      # seasonal factor
gdp_i <- gdp_lt_sa$final$series[, "i"]      # irregular component

# Create data frame with final data
data_sa <- data.frame(
  time = as.yearqtr(time(gdp_nsa)),
  gdp_nsa = as.numeric(gdp_nsa),
  gdp_sa = as.numeric(gdp_sa),
  gdp_t = as.numeric(gdp_t),
  gdp_s = as.numeric(gdp_s),
  gdp_i = as.numeric(gdp_i)
)
```

```
plot(gdp_s,
     main = "Seasonal component",
     ylab = "",
     xlab = "",
     lwd = 2)

plot(gdp_i,
     main = "Irregular component",
     ylab = "",
     xlab = "",
     lwd = 2)
```

```
# Visualizing results

# Set up 3 rows × 2 columns layout
par(mfrow = c(1, 1)     # 3 rows, 2 columns
)

plot(gdp_nsa,
     main = "Original data",
     ylab = "",
     xlab = "",
     lwd = 2)

plot(gdp_sa,
     main = "Seasonally adjusted data",
     ylab = "",
     xlab = "",
     lwd = 2)

plot(gdp_t,
     main = "Trend-cycle component",
     ylab = "",
     xlab = "",
     lwd = 2)
```

Seasonal adjustment.
Step 2. Adjustment: R code X-13

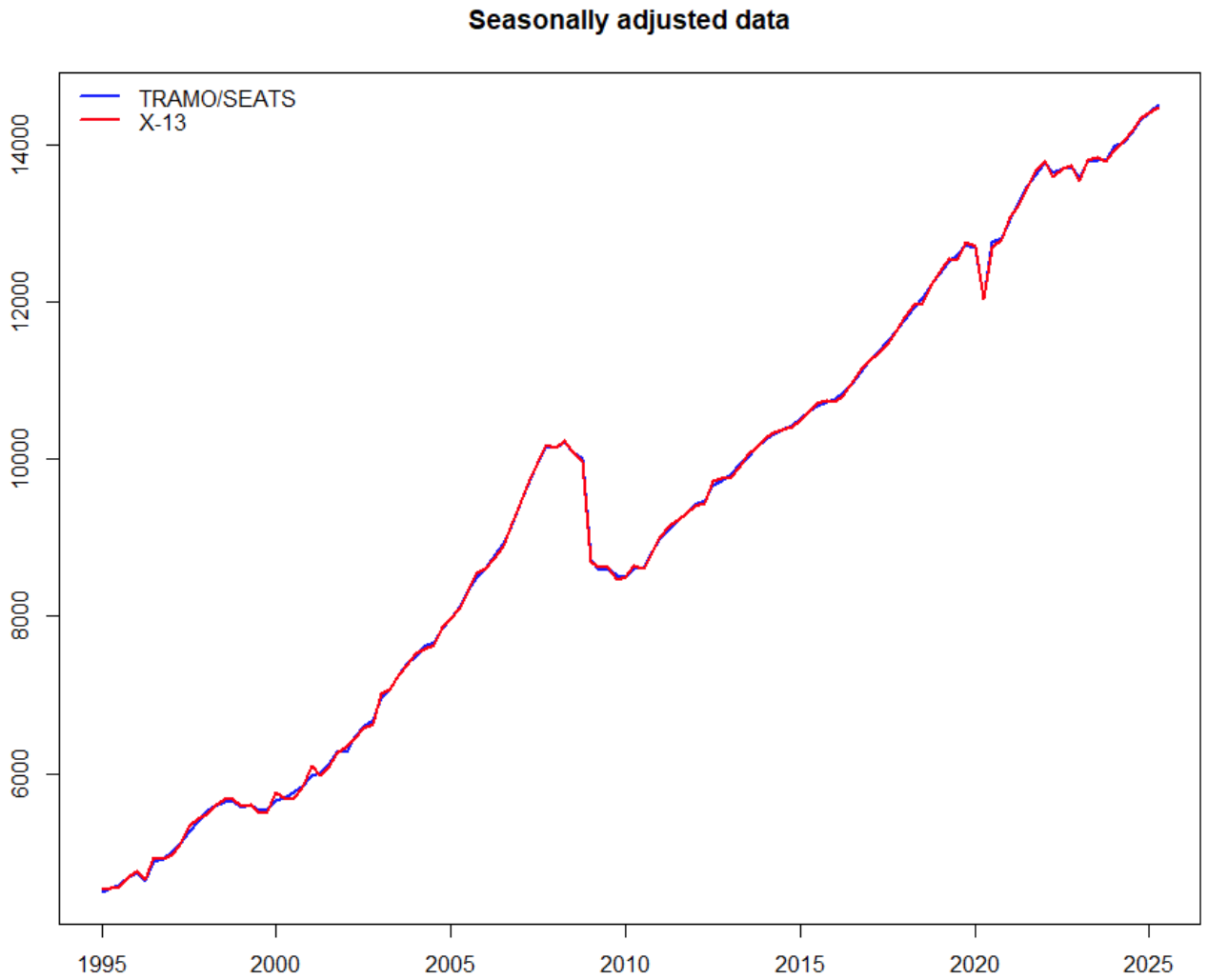
```
# Seasonal Adjustment: X-13
gdp_lt_sa_x13 <- x13(
  gdp_lt_ts, spec = c("RSA4c"))

# Results

plot(
  gdp_lt_sa_x13,
  forecast = FALSE,
  type_chart = c("sa-trend", "cal-seas-irr"),
  caption = c(`sa-trend` = "Original and seasonally adjusted data", `cal-seas-irr` =
                "Seasonal & Irregular Components"),
  ask = FALSE
)

gdp_nsa_x13 <- gdp_lt_sa_x13$final$series[, "y"]    # original time series
gdp_sa_x13 <- gdp_lt_sa_x13$final$series[, "sa"]    # seasonally adjusted time series
gdp_t_x13 <- gdp_lt_sa_x13$final$series[, "t"]      # trend-cycle components
gdp_s_x13 <- gdp_lt_sa_x13$final$series[, "s"]      # seasonal factor
gdp_i_x13 <- gdp_lt_sa_x13$final$series[, "i"]      # irregular component
```

Seasonal adjustment.
Step 2. Adjustment: TRAMO/SEATS vs. X-13

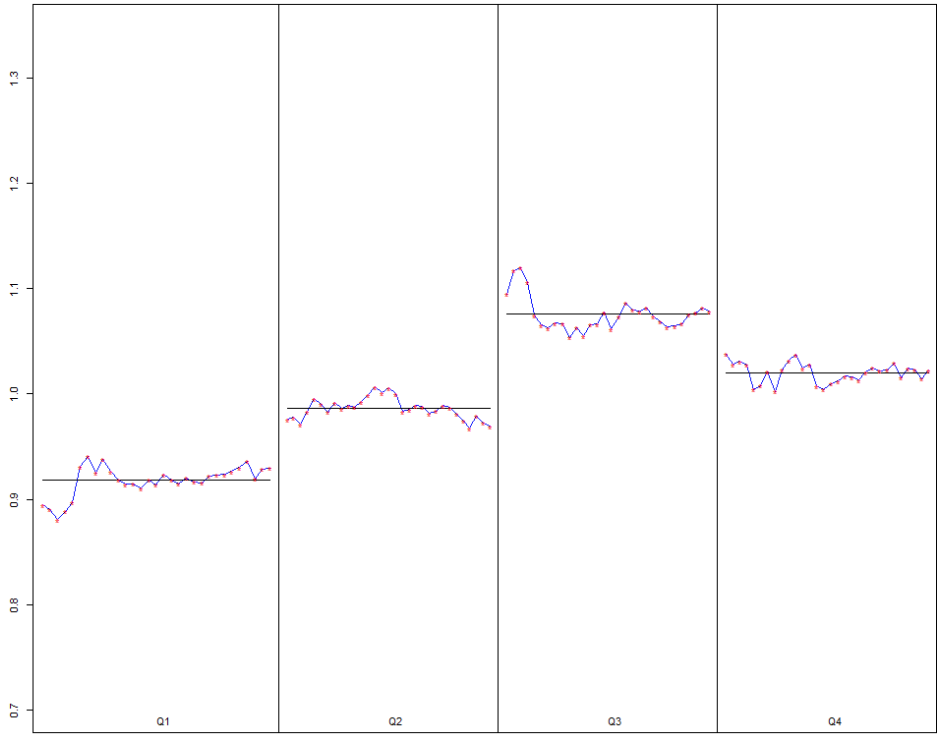


Seasonal adjustment.
Step 3. Quality check: S-I ratio

The **S-I ratio** chart presents the seasonal-irregular (S-I) component and the seasonal factors for each of the periods in the time series (months or quarters). **Blue curves** represent the ratio of the original time series to the trend-cycle component, or, equivalently, the product of the seasonal factor and the irregular component. **Black straight** lines represent the mean seasonal factor for each period.

```
# S-I ratio
plot(gdp_lt_sa$decomposition)
```

In our case we don't see significant changes in seasonality



Seasonal adjustment.
Step 3. Quality check: Estimated model, outliers

```
> gdp_lt_sa$regarima
y = regression model + arima (1, 0, 0, 0, 1, 0)
Log-transformation: yes
Coefficients:
       Estimate Std. Error
Phi(1)  -0.8191      0.054

             Estimate Std. Error
Mean          0.04318      0.009
LS (I-2009)  -0.13859      0.014
AO (II-2020) -0.05610      0.010
AO (II-1996) -0.03782      0.010

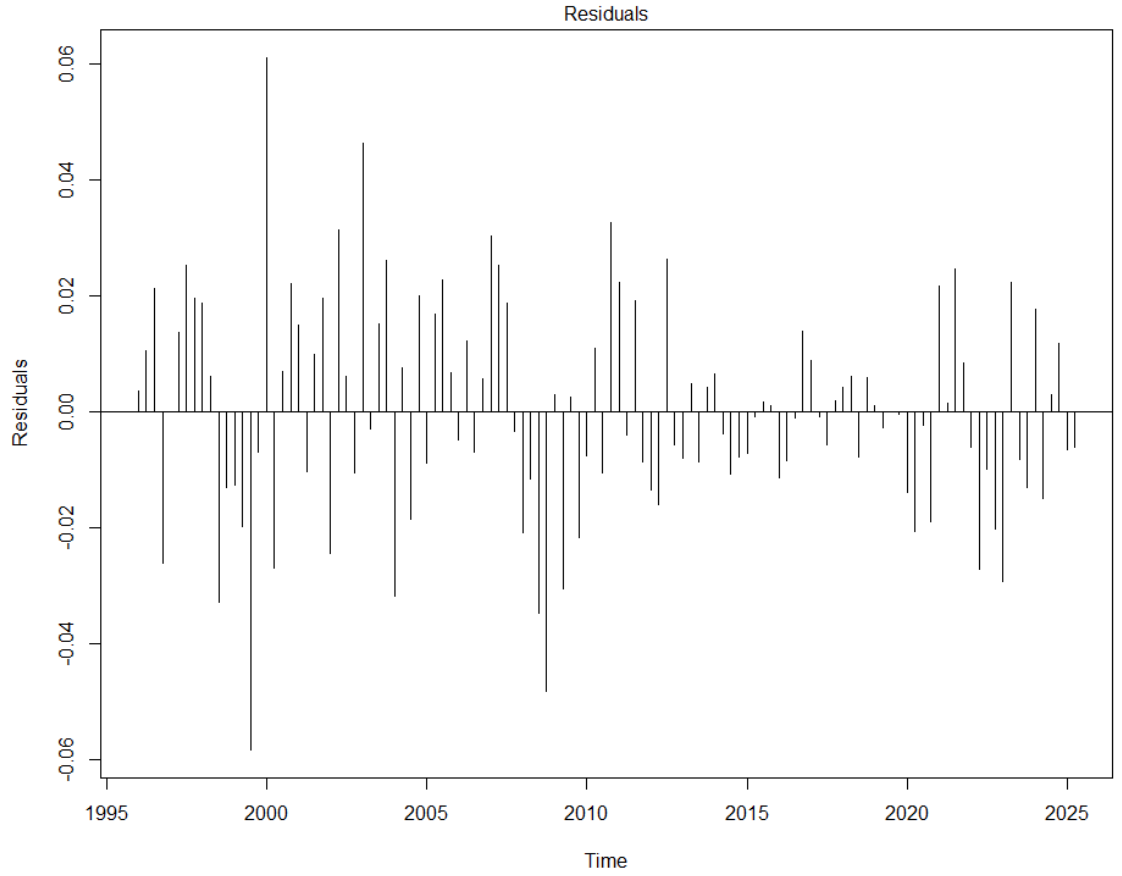

Residual standard error: 0.0183 on 112 degrees of freedom
Log likelihood = 304.1, aic =  1553 aicc =  1554, bic(corrected for length) =  -7.8
```

In general, the automatic model identification should produce satisfactory results, and it should not be needed to manually define the model.

Seasonal adjustment.
Step 3. Quality check: Residuals (1)

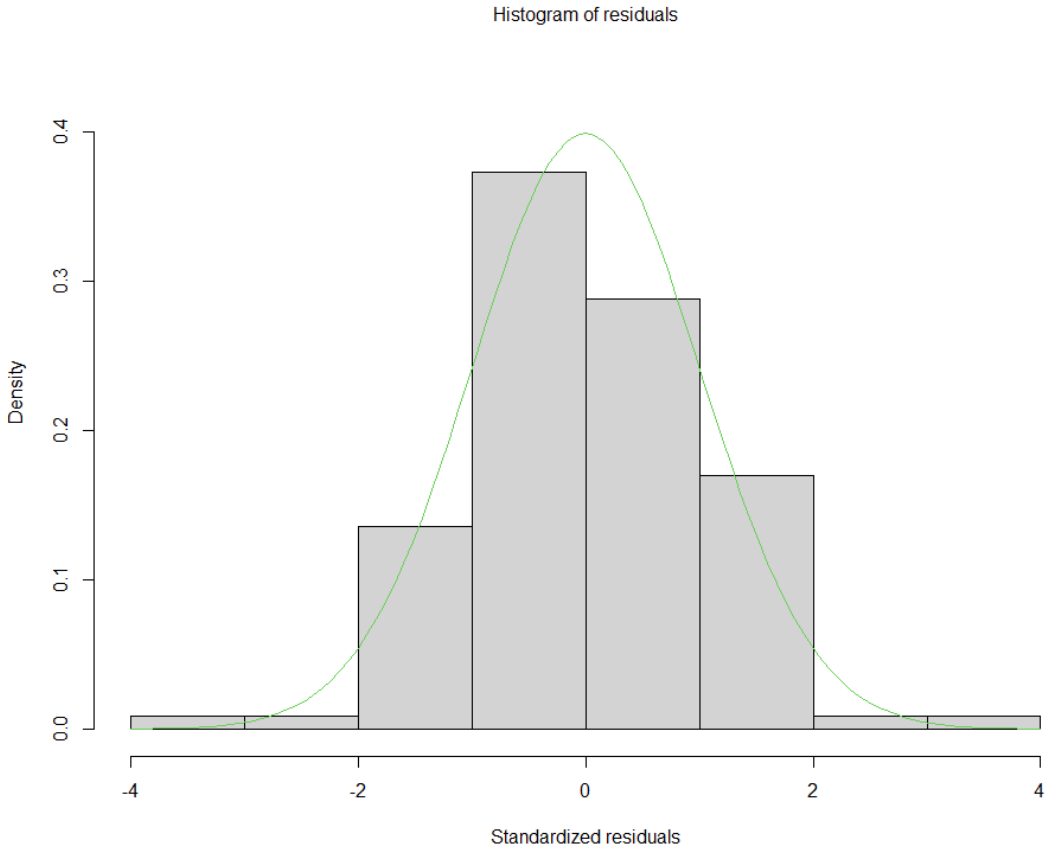
```
plot(gdp_lt_sa$regarima)
plot(gdp_lt_sa$regarima, which = 7)
```

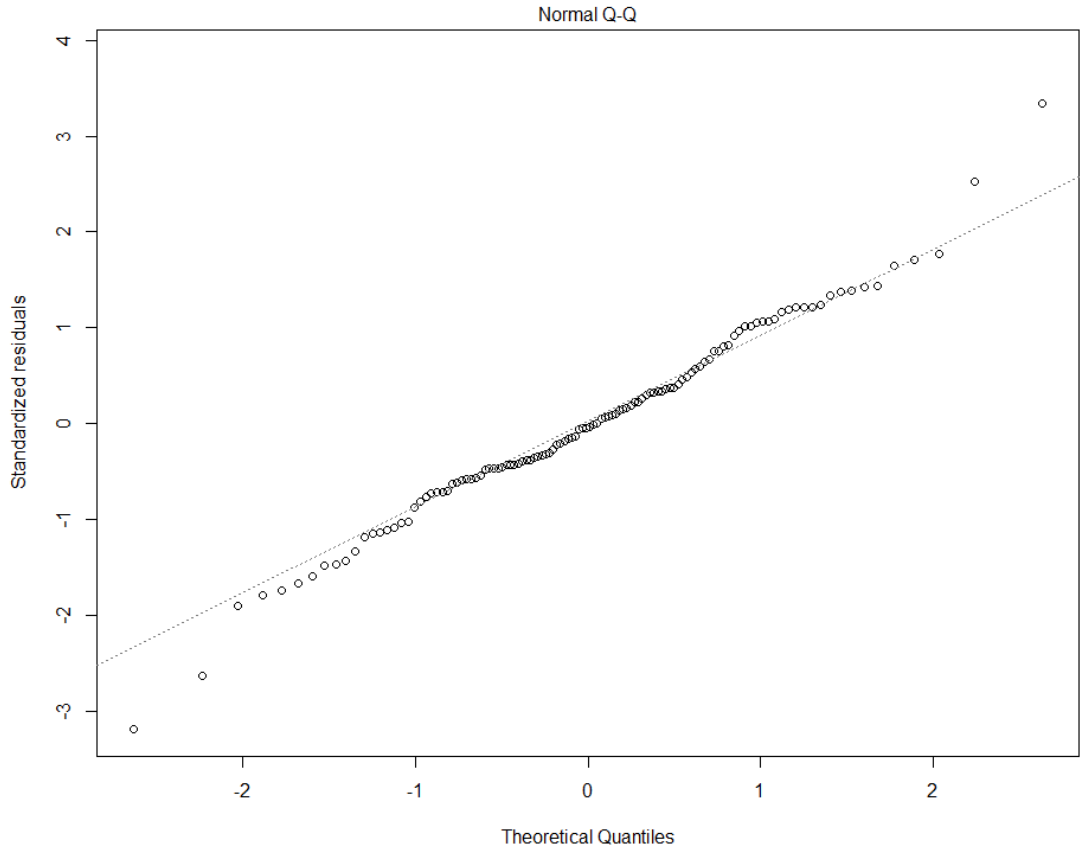
The residuals should be independent and random and follow the normal distribution with a zero mean.



Seasonal adjustment.
Step 3. Quality check: Residuals (2)

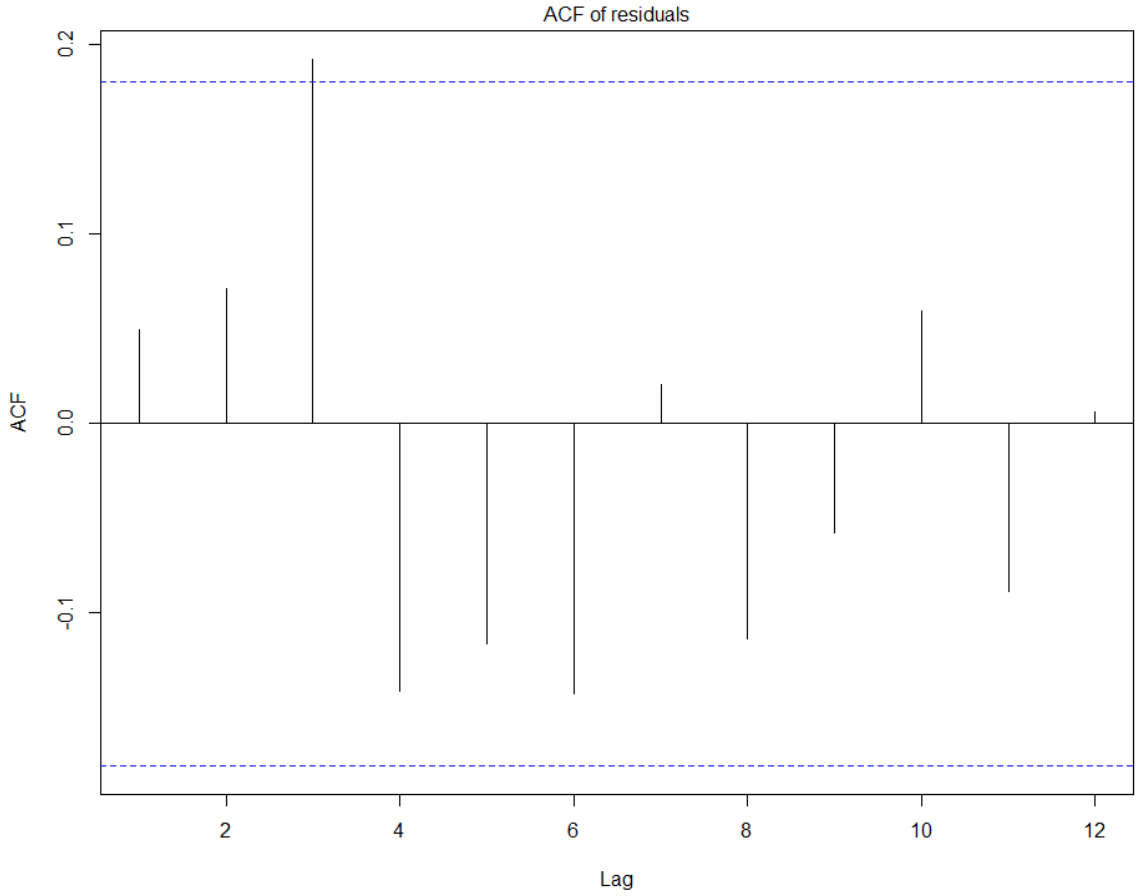
Residuals have zero mean and close to normal distribution

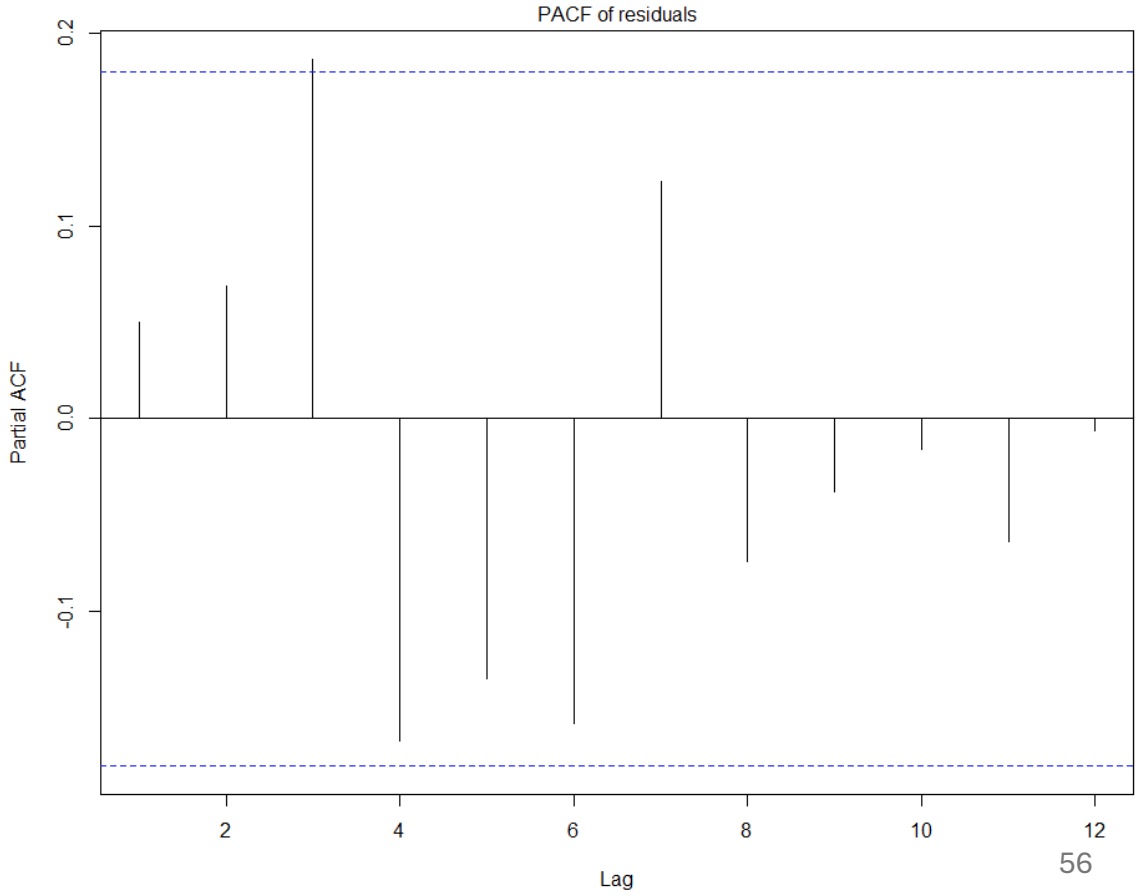




Seasonal adjustment.
Step 3. Quality check: Residuals (3)

Based on autocorrelations and partial autocorrelations, we can conclude that residuals are independent – don't depend on the lagged values.





Seasonal adjustment.
Step 3. Quality check: Seasonality in time series

```
> gdp_lt_sa$diagnostics
Relative contribution of the components to the stationary
portion of the variance in the original series,
after the removal of the long term trend
 Trend computed by Hodrick-Prescott filter (cycle length = 8.0 years)
           Component
 Cycle         4.310
 Seasonal     36.667
 Irregular     0.089
 TD & Hol.     0.000
 Others       51.293
 Total        92.358

Combined test in the entire series
 Non parametric tests for stable seasonality
                                                          P.value
   Kruskall-Wallis test                                         0
   Test for the presence of seasonality assuming stability      0
   Evolutive seasonality test                                   0

 Identifiable seasonality present
```

In our case identifiable seasonality presents.

Seasonal adjustment.
Step 3. Quality check: Residual seasonality

Residual seasonality tests

	P.value
qs test on sa	1.000
qs test on i	1.000
f-test on sa (seasonal dummies)	1.000
f-test on i (seasonal dummies)	1.000
Residual seasonality (entire series)	1.000
Residual seasonality (last 3 years)	0.823
f-test on sa (td)	0.644
f-test on i (td)	0.003

The results are fully in line with the expectations. There is no residual seasonality in the entire duration of the series and for the last three years of the series.

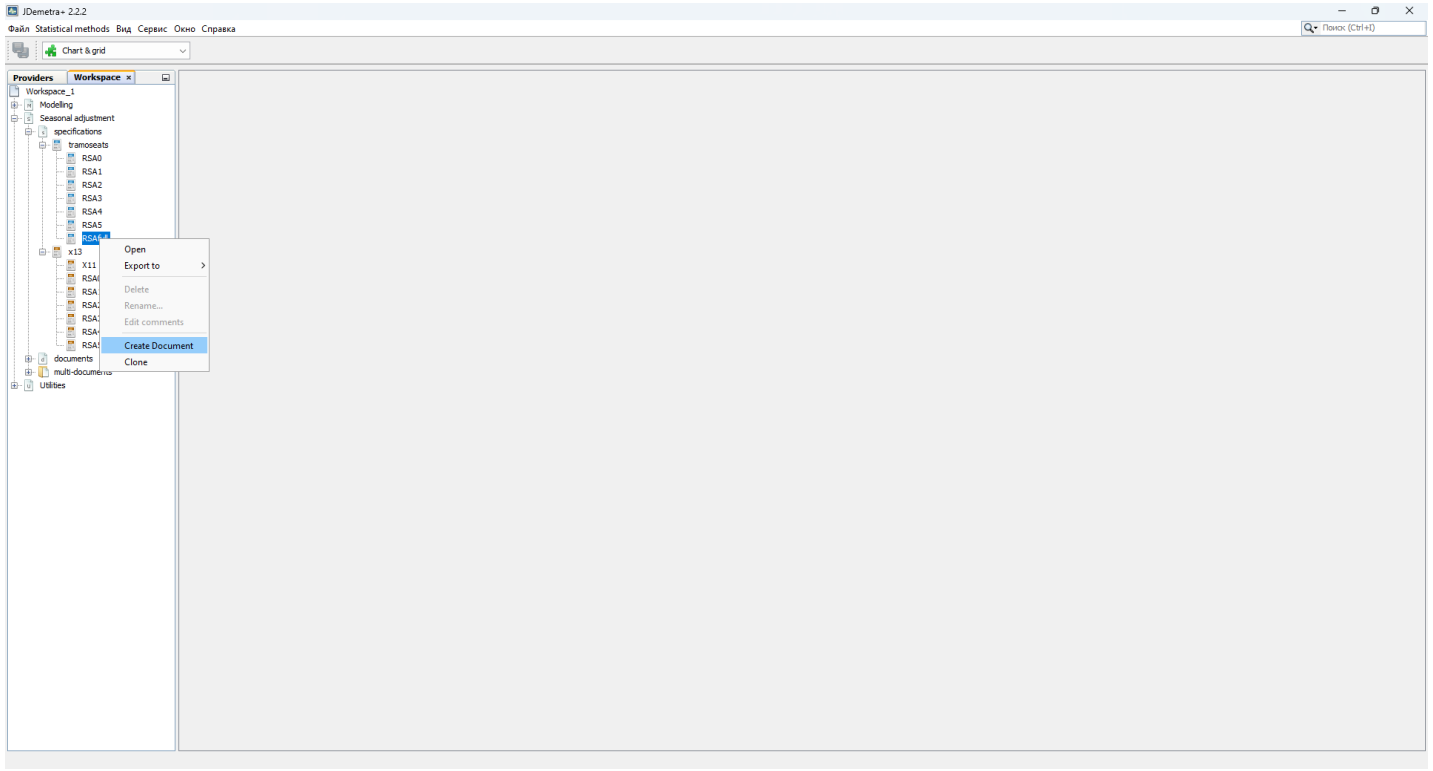
A4. Seasonal adjustment in JDemetra+

Creating a document

1. Open JDemetra+.

2. Choose the method of seasonal adjustment:

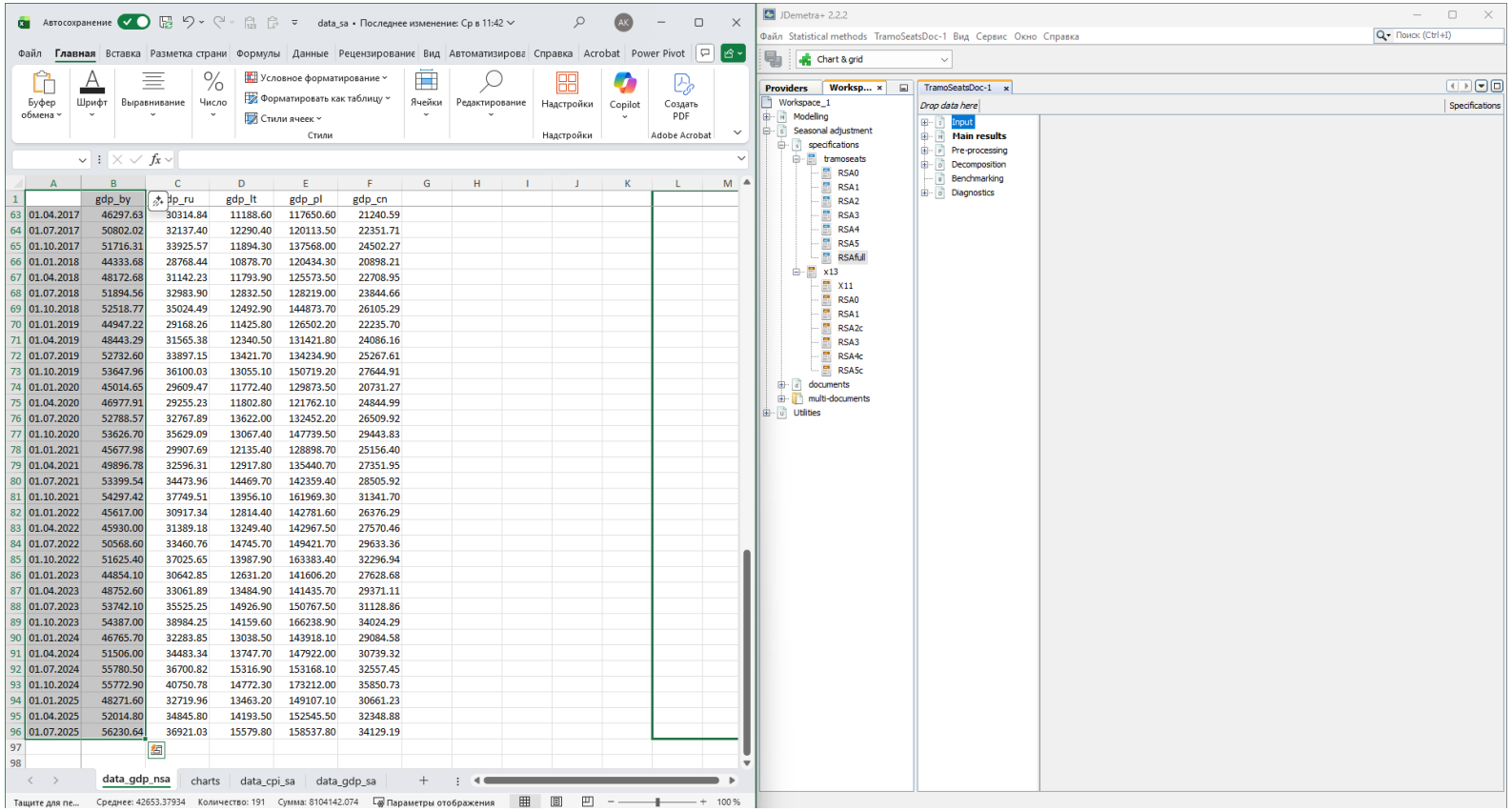
Workspace / Seasonal adjustment / specifications / tramoseats (or x13) / your choice (RSAfull or RSA4c are suggested) / right click – Create document



Importing data

3. Importing data from Excel to JDemetra+.

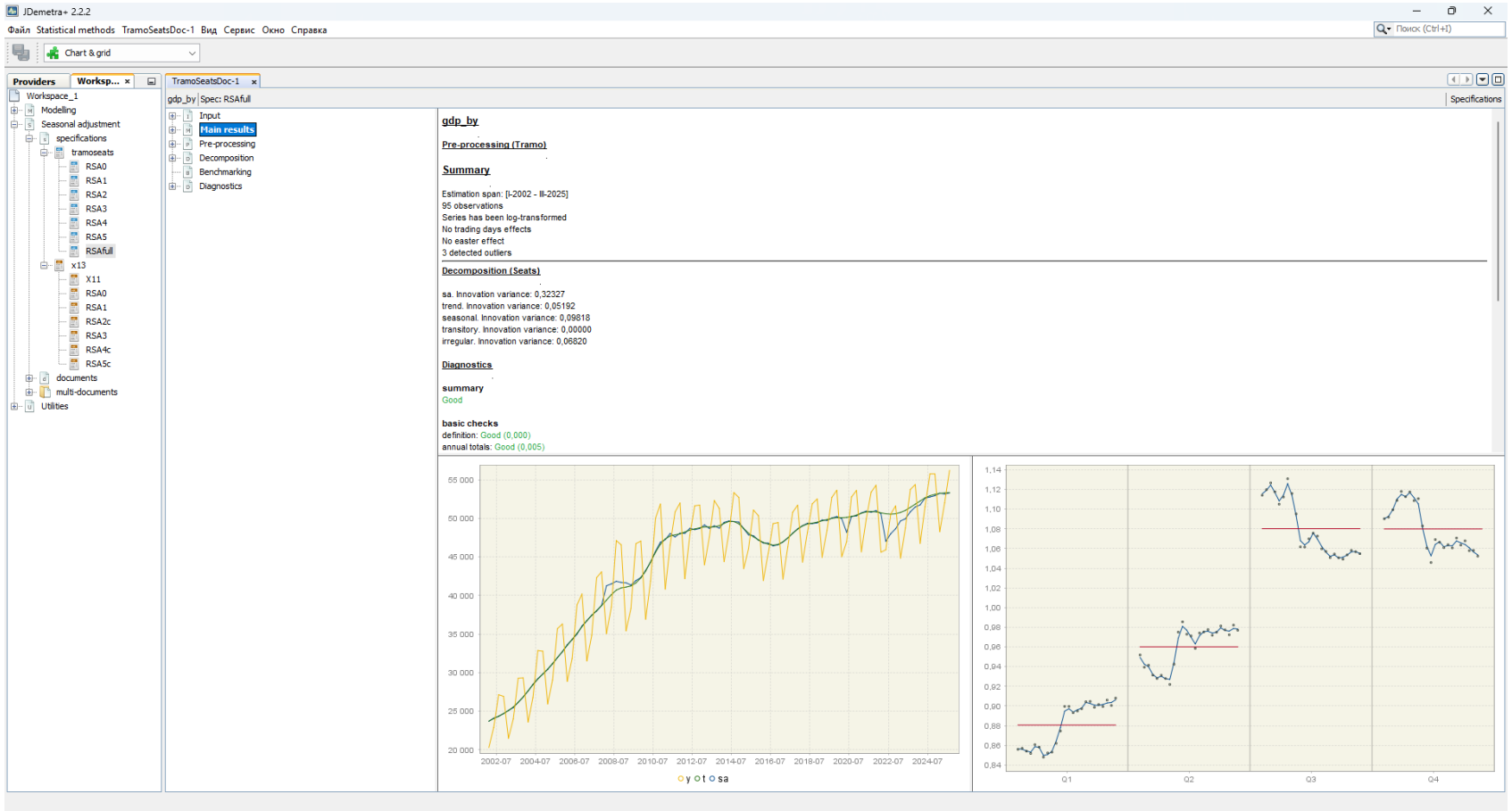
Select your data in Excel (select two columns – with time periods and with data, along with the headers – for time periods it is empty). Drag the selected data to JDemetra+ space (select the “Input” line before dragging).



Results

4. Check the seasonal adjustment results.

Look at “Main results”.

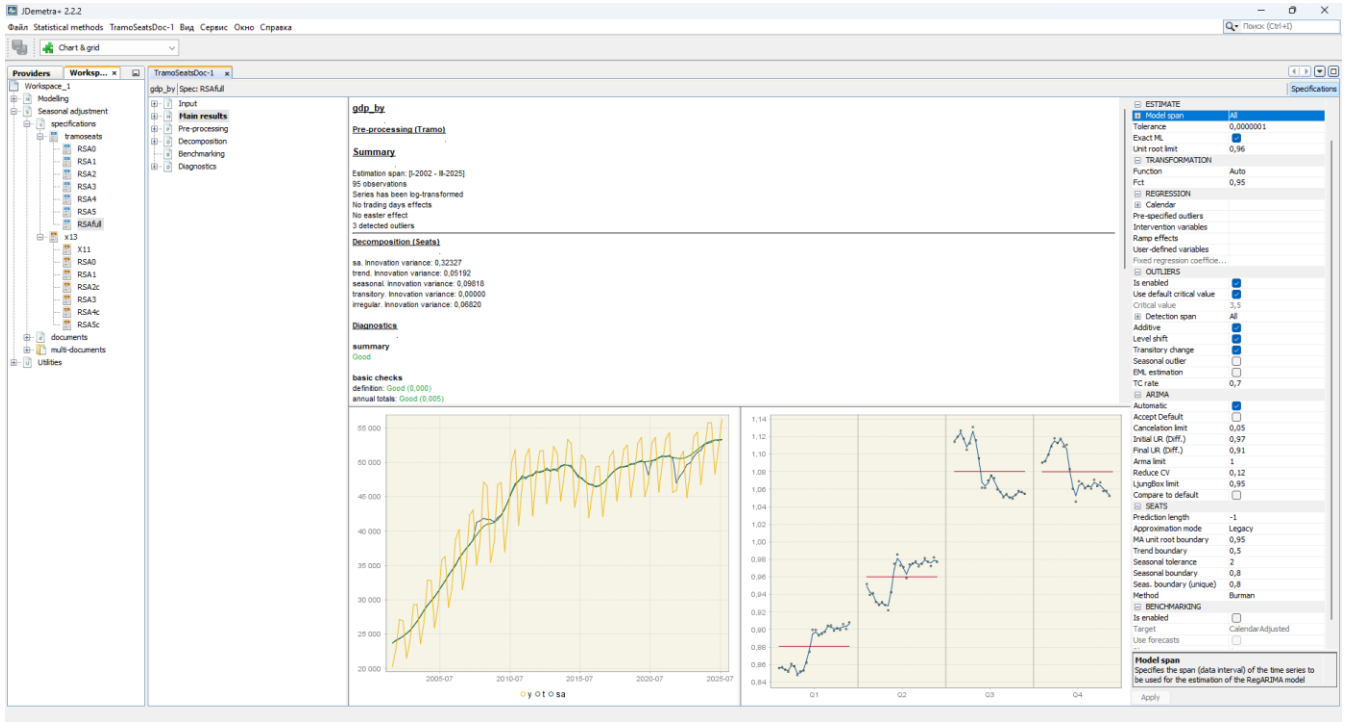


Quality of the seasonal adjustment (1)

5. Assess the quality of the seasonal adjustment.

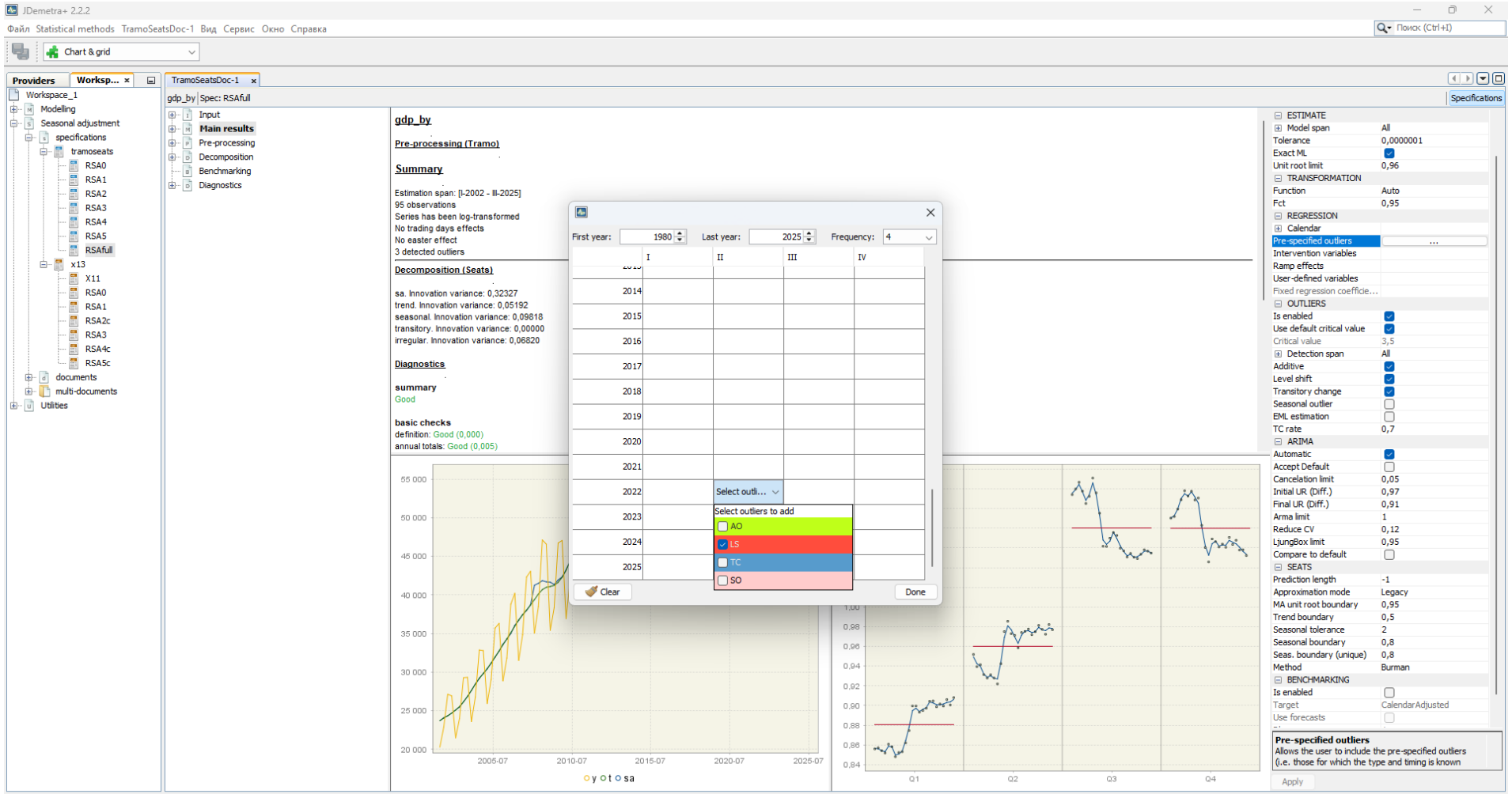
Check the “Diagnostics” from the “Main results”. If the summary is **good** or **uncertain**, than the results are satisfactory. If the summary is **severe**, than your seasonal adjustment is not good. Try to 1) shorten the time period to account for breaks; 2) identify outliers; 3) include the leap year.

6. To change time period, click on the “Specifications” and change “Series span”. Click “Apply”.



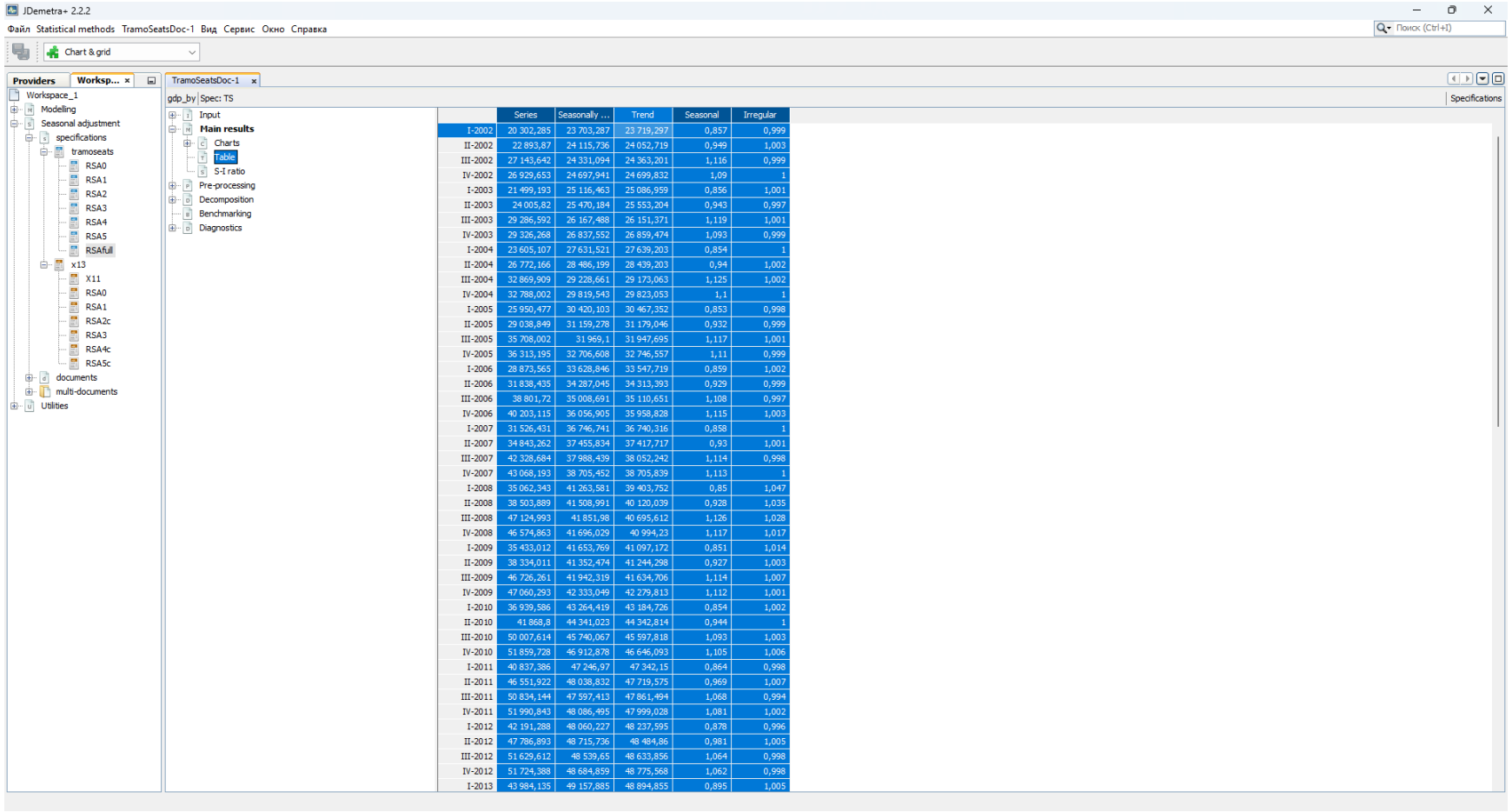
Quality of the seasonal adjustment (2)

7. To include outliers, click on the “Specifications” and select “Pre-specified outliers”.



Saving data

8. Your seasonally adjusted data are stored in “Main results / Table”. Click on the top left empty sell to select the whole table. Copy it and paste it to Excel.



	Series	Seasonally ...	Trend	Seasonal	Irregular
I-2002	20 302,285	23 703,287	23 719,297	0,857	0,999
II-2002	22 893,87	24 115,736	24 052,719	0,949	1,003
III-2002	27 143,642	24 331,094	24 363,201	1,116	0,999
IV-2002	26 929,653	24 697,941	24 699,832	1,09	1
I-2003	21 499,193	25 116,463	25 086,959	0,856	1,001
II-2003	24 005,82	25 470,184	25 553,204	0,943	0,997
III-2003	29 286,592	26 167,488	26 151,371	1,119	1,001
IV-2003	29 326,268	26 837,552	26 859,474	1,093	0,999
I-2004	23 605,107	27 631,521	27 639,203	0,854	1
II-2004	26 772,166	28 486,199	28 439,203	0,94	1,002
III-2004	32 869,909	29 228,661	29 173,063	1,125	1,002
IV-2004	32 788,002	29 819,543	29 823,053	1,1	1
I-2005	25 950,477	30 420,103	30 467,352	0,853	0,998
II-2005	29 038,849	31 159,278	31 179,046	0,932	0,999
III-2005	35 708,002	31 969,1	31 947,695	1,117	1,001
IV-2005	36 313,195	32 706,608	32 746,557	1,11	0,999
I-2006	28 873,565	33 628,846	33 547,719	0,859	1,002
II-2006	31 838,435	34 287,045	34 313,393	0,929	0,999
III-2006	38 801,72	35 008,691	35 110,651	1,108	0,997
IV-2006	40 203,115	36 056,905	35 958,828	1,115	1,003
I-2007	31 526,431	36 746,741	36 740,316	0,858	1
II-2007	34 843,262	37 455,834	37 417,717	0,93	1,001
III-2007	42 328,684	37 988,439	38 052,242	1,114	0,998
IV-2007	43 068,193	38 705,452	38 705,839	1,113	1
I-2008	35 062,343	41 263,581	39 403,752	0,85	1,047
II-2008	38 503,889	41 508,991	40 120,039	0,928	1,035
III-2008	47 124,993	41 851,98	40 695,612	1,126	1,028
IV-2008	46 574,863	41 696,029	40 994,23	1,117	1,017
I-2009	35 433,012	41 653,769	41 097,172	0,851	1,014
II-2009	38 334,011	41 352,474	41 244,298	0,927	1,003
III-2009	46 726,261	41 942,319	41 634,706	1,114	1,007
IV-2009	47 060,293	42 333,049	42 279,813	1,112	1,001
I-2010	36 939,586	43 264,419	43 184,726	0,854	1,002
II-2010	41 868,8	44 341,023	44 342,814	0,944	1
III-2010	50 007,614	45 740,067	45 597,818	1,093	1,003
IV-2010	51 859,728	46 912,878	46 646,093	1,105	1,006
I-2011	40 837,386	47 246,97	47 342,15	0,864	0,998
II-2011	46 551,922	48 038,832	47 719,575	0,969	1,007
III-2011	50 834,144	47 597,413	47 861,494	1,068	0,994
IV-2011	51 990,843	48 086,495	47 999,028	1,081	1,002
I-2012	42 191,288	48 060,227	48 237,595	0,878	0,996
II-2012	47 786,893	48 715,736	48 484,86	0,981	1,005
III-2012	51 629,612	48 539,65	48 633,856	1,064	0,998
IV-2012	51 724,388	48 684,859	48 775,568	1,062	0,998
I-2013	43 984,135	49 157,885	48 894,855	0,895	1,005